

How contraflow enhances clearance time during assisted mass evacuation – A case study exploring the Australian 2013–14 Gippsland bushfires

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ABSTRACT

Evacuation during a catastrophic disaster is a crucial operation that needs to be appropriately managed and is of more importance when considering the elderly and people with disabilities. The uncertain and unpredictable nature of disasters can cause long-term repercussions, especially in traffic congestion. This study presents a mathematical model to formulate traffic balance for regular and assisted evacuation (that is, disabled and the elderly) whilst considering traffic congestion in evacuation clearance time by applying contraflow. A Branch and Price (B&P) related approach is developed to help solve the proposed model in large-size problems. The presented algorithm is applied to a case study of Australia's 2013–14 bushfires in Gippsland, located in the eastern part of Victoria. A variation test is performed to evaluate the robustness of results generated by the developed model. Results indicate that the participation percentage of edges is different based on their location, capacity, and sustainability of blockage. The edges' capacity influences the evacuated population most compared to route capacity and time window. The output of this approach enables authorities to improve the resilience of communities by making optimal strategic and operational decisions for enhancing an evacuation response as well as influencing appropriate policies.

1. Introduction

Critical infrastructure services refer to the systems or networks that provide for the circulation of people, services, and information, upon which health, safety, comfort, and economic activity depend (Johnston et al., 2006). These services are crucial for regional and remote communities to maintain life and sustain communities during times of crisis. Critical infrastructure, such as transportation networks, is often viewed as a national and regional issue, depending on the crisis. These are also a significant focus for the United Nations Office for disaster risk reduction providing our most basic needs to support our well-being and development (Wannous and Velasquez, 2017). Infrastructure systems are under increasing pressure to deliver resilient and reliable services in this time of rapid climate change and intensifying natural hazards. Moreover, unforeseen consequences can result from increasing interconnections between systems and networks, leading to greater potential for cascading disasters in which the loss of one system can trigger failures in others (Matter and Strength, 2018; Gissing et al., 2022).

Evacuation is the “movement of people threatened by a hazard to a safer location and typically, their eventual safe and timely return” (TasmanianGovernment, 2018, pp. 6) and is widely accepted as a protective action in emergencies such as bushfires, floods,

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blizzards, etc., regardless of the hazard for improving the resilience of communities (Sorensen et al., 2004). The efficient scheduling of evacuees in emergency conditions has been a long debatable challenge (He et al., 2015). For example, when Hurricane Rita struck the Gulf of Mexico, a large number of people evacuated at the same time, which led to a long traffic jam (100 miles) on Route 45 exiting Texas (Kim et al., 2008) and left many stranded and out of petrol. Evacuation missions can be alleviated if an effective plan has been prepared. Such plans allow for efficient evacuation and reduced losses experienced (He et al., 2015). Reliance on critical infrastructure, network segments, and essential services are key vulnerabilities globally (Afkham et al., 2022). Identifying and prioritising accessible routes of the transportation network aids in minimising conflicts that may occur at intersections (Bretschneider and Kimms, 2011). However, identifying and minimising congestion is a challenge for any evacuation plan, thus increasing the risk and limiting the vacating of the affected region/s (U.S. House, 2006). Evacuation is typically known as the safest option. However, some people in the affected region/s prefer to stay at home since they are not aware of the magnitude of the risk or fail to take the risk seriously until it is late (Shahparvari et al., 2015a, 2016, 2017; Shahparvari and Abbasi, 2017). Not only is this a practical issue that needs attention, but one that needs action through policy, influencing a range of stakeholders, such as emergency authorities, police, and various levels of government.

In Australia, the context of this study, the impact of bushfires between 1967 and 1999 has been estimated to be worth approximately AUD 2.5 billion with 233 fatalities (Shahparvari et al., 2021). The estimates of 2009 Black Saturday bushfires were estimated to cause AUD 4.4 billion in damages and 173 fatalities (Shahparvari et al., 2019; Shahparvari and Bodaghi, 2018). More recently, “Black Summer bushfires” in Australia (September 2019–March 2020) burnt an estimated 18.6 million hectares, destroyed over 5,900 buildings (including 2,779 homes), and killed at least 34 people (Tiernan and O’Mallon, 2020). An estimated one billion animals were killed, and some endangered species were driven to the verge of extinction (Green, 2020). At its peak, the air quality dropped to hazardous levels, which made evacuation even more dangerous (Longman, 2020). The smoke had extended approximately 11,000 kilometres across the South Pacific Ocean to Chile and Argentina (Lee, 2020). Ten thousand occupants had been urged to flee following the declaration of a state of emergency by the government, while four thousand were stranded in the Murrumbidgee coastal township as the arterial road had been blocked by the fire on New Year’s Eve (Martin, 2019). Black summer bushfires are estimated to cost more than Black Saturday fires in 2009 (\$4.4 billion) (Butler, 2020). The Australian tourism industry has experienced a dramatic fall of more than \$1 billion as well (Huf and Mclean, 2020). In this vein, the Royal Commission into National Natural Disaster Arrangements, and its associated recommendations, focus on understanding and reducing the risks to essential services arising from emergencies and disasters caused by natural hazards in Australia (Binskin et al., 2020).

In a broader context, catastrophic events such as heatwaves, volcanic eruptions, wildfires, hurricanes, tornadoes, tsunamis, and earthquakes, as well as human-made disasters such as fires, deforestation, nuclear activities, and industrial accidents, have accentuated the importance of one of the most significant problems in transportation, evacuation (Moshtagh et al., 2018). This problem is not isolated to just one location and usually has a domino effect, impacting the environment, people’s livelihood, economies, and communities (Kamiyama et al., 2009). The rationale of this study originates from recommendations of the “Victorian bushfires royal commission” (Teague et al., 2010). The “royal commission into national natural disaster arrangements” (Binskin et al., 2020) reports which both emphasise “the necessity of urgent development and improvement of emergency evacuation components such as identification of neighbourhood safer places and potential shelters” (Teague et al., 2010, pp. 20–21) to improve the entire process to leverage the loss/death toll. Thus, there is a need for appropriate planning for efficiently scheduling evacuations from impacted areas with minimal interruption or delay. This crisis management is made even more difficult for people who may be disabled, the elderly, or people who live alone.

This is a global issue that community leaders and authorities have to resolve by identifying and developing evacuation plans to safe locations promptly when confronted by natural or human-made disasters (Nakanishi et al., 2019). By developing appropriate evacuation plans, we are also able to lessen the impact on community resilience by minimising “stress and crisis-induced transient periods of dysfunction” (Norris et al., 2008) (p. 144) that disasters bring about. However, this is more complicated for older people and people with disabilities, who require special attention in supporting and evacuating disaster-hit areas. (Tanaka, 2013) presented a case regarding one 17-year-old male, identifying that despite living in a single-storied building about 500 m from the coast, to evacuate him would have “required at least two people to carry him and one to carry his medical equipment: a total of three people” (p. 210). Further, in this context, (Tanaka, 2013) estimated that twice as many disabled individuals were casualties in the Japanese earthquake in 2011. The disabled individuals in this study were casualties of a catastrophic earthquake that triggered powerful tsunami waves. The impact of the event and the short warning time, combined with evacuation issues, led to their inability to evacuate. This study highlights the difficulty of evacuating disabled individuals, requiring multiple people to carry them and their medical equipment. Therefore, designing a tactical and efficient scheduling system to transport the affected population (particularly the elderly and people with disabilities) to safe shelters in the least amount of time and with little congestion is paramount to minimise the impact of these disasters on people’s livelihood, economies, and communities. This scheduling system also needs to be enacted through policy and the inclusion of all associated stakeholders. In addition, Fountain et al. (2019) investigated the stakeholder consultation to define research priorities about the elderly (i.e., older adults) and natural disasters in Australia and compared their study and current research literature in this area. Similarly, Shahparvari et al. (2015b) found through multi-objective analytics from three bushfire scenarios that older adults pose challenges for emergency authorities in the late evacuation stage, which involves timely evacuation, shelter assignment, and routing.

Free-flow models have typically been used in evacuation planning research. However, the model developed and presented in this paper does not follow traditional evacuation methodologies focusing on evacuation tracks allocated to regions that should be evacuated (Pillac et al., 2016). Instead, this study focuses on how residents evacuate during emergencies with time constraints and traffic congestion. Furthermore, little research focuses on people with disabilities and the elderly and their evacuation patterns.

Therefore, this research aims to design a practical, tactical and efficient scheduling system to transport people from areas affected by disaster to a safe shelter with minimal delays in both time and congestion, supporting all emergency authorities appropriately.

In the context of bushfire evacuations, improving resilience and informing operational decisions can involve a range of strategies and actions. Some potential ways that the research output of the paper could contribute to these goals include:

- **Optimising evacuation routes:** The modelling approach presented in the paper can help emergency management officials identify optimal evacuation routes that minimise congestion and maximise the number of evacuees that can safely and quickly exit the affected area. This information can be used to develop evacuation plans and improve the efficiency of evacuation operations.
- **Enhancing communication and information dissemination:** The paper highlights the importance of providing accurate and up-to-date information to evacuees during bushfire events. Emergency management officials can use the insights from the research to develop communication strategies that effectively inform the public about the status of the fire, evacuation orders, and available resources.
- **Assessing the impact of different evacuation scenarios:** The modelling approach presented in the paper can help emergency management officials evaluate the potential impact of different evacuation scenarios, such as evacuating high-risk areas or using contraflow lanes. This information can inform decision-making and help officials make more informed and effective operational decisions.
- **Assisting vulnerable populations:** The research can be used to assist people at higher risk during bushfire evacuations, such as individuals with limited mobility or those living in remote areas. This information can be used to develop targeted evacuation plans and ensure these populations can access the necessary resources and support. The research output of the paper, hence, can contribute to improving the resilience of communities during bushfire evacuations by providing emergency management officials with the tools and insights necessary to make informed and effective operational decisions.

The main contributions of this study are as follows:

- A novel integrated assisted mass evacuation model is developed.
- Evacuation of elderly people and people with disabilities are investigated in this study.
- An Accelerated Branch & Price algorithm is proposed to deal with large-scale evacuation problems.
- Contraflow is incorporated in the developed evacuation model to enhance the network's capacity and minimise the total clearance time.

The rest of the paper is organised as follows: the problem background is further discussed in Section 2. Section 4 presents the model formulation. Section 5 as the solution approach develops a B&P algorithm for the proposed model. Section 7 provides numerical experiments. Section 8 describes the case study, results, and analysis. Section 9 provides managerial insights for the decision-makers. Finally, Section 10 concludes the paper.

2. Problem background

Commonly, evacuation has been used to describe a move from at-risk areas to safer areas (Southworth, 1991). Depending on its scale and the level of authority control, evacuation may differ in terms of its properties (Stepanov and Smith, 2009). Whilst the existing literature on emergency evacuation is comprehensive, we are still experiencing significant challenges that require resolution. Time-related networks have been developed in most evacuation scheduling studies to examine evacuation scheduling (Pillac et al., 2016). For example, He et al. (2015) proposed an earliest arrival flow formulation for determining the allocation of dynamic resources in the evacuation schedule. The contraflow scheme has also been adopted to alleviate congestion (Pillac et al., 2016). Kim et al. (2008) set a contraflow network configuration to minimise evacuation time for each edge. In addition, Kamiyama (2019) designed a dynamic graph with a single sink vertex, supplied with all vertexes except the sink vertex. The self-evacuation and bus evacuation have both been examined within this stream. Similarly, Dikas and Minis (2016) investigated the evacuation of residents and the transport of casualties afterwards.

Research indicates that many intersecting social factors affect people's evacuation decisions, including engagement with and observations of others (Du et al., 2017), which leads to complex feedbacks. In terms of individuals' thought processes, risk perception and motivation are two important components of evacuation decision-making (Kinsey et al., 2019), though high-risk perception does not automatically increase preparedness (Wachinger et al., 2013). Individuals' subjective assessments of the severity, likelihood, and potential consequences of a hazard – that is, their perception of risk – are influenced by the proximity of the hazard, past experiences, trust in information sources, social networks, and cognitive biases (Dash and Gladwin, 2007; Lechner and Rouleau, 2019). Individual's motivation to evacuate is shaped not only by a desire to protect themselves and loved ones but the perceived effectiveness of a given evacuation option including the perceived safety, or lack thereof, of shelter options. People draw on various sources of information – such as official announcements, media, and observation of and messages from their social networks – to assess a given threat situation and determine the desirability and feasibility of evacuation options. Effective communication strategies, clear warnings, and instructions from authorities can positively influence decision-making (Kinsey et al., 2019). Whether people respond to evacuation orders depends in part on their trust in authorities (Dash, 2002; Kinsey et al., 2019), underlining the importance of sustained community engagement by authorities prior to disasters (Satizabal et al., 2022). Social encouragement

and support can strongly influence evacuation decision-making, especially for groups requiring assistance (Du et al., 2017; Na and Grace, 2022).

The decision-making process during evacuations is often complex, dynamic and highly iterative (Kinsey et al., 2019). It involves weighing potential costs and benefits, evaluating multiple options, and adapting decisions based on changing circumstances. Researchers have used decision-making models, such as the Protection Motivation Theory (PMT) and the Theory of Planned Behaviour (TPB) to try to understand evacuation decision-making processes (Marzouk and Mohamed, 2019). Three main groups can be distinguished on the basis of their evacuation decisions (Cova et al., 2011): 1. Voluntary early evacuees (who contribute to what they call 'background traffic'); 2. Compliant evacuees who follow warnings and detailed evacuation plans, including assisted evacuation (participating in 'mass evacuation'); and (3) Non-evacuees who choose to shelter in place, either in contravention of a mandatory evacuation order or in accordance with policies such as Australia's 'Prepare, Stay and Defend or Leave Early' policy. Subsequent research, however, underlines that evacuation is not a simple, zero sum game: people oscillate in their decisions and actions, with some attempting, reversing and reattempting evacuation multiple times as disaster situations unfold (Kuligowski, 2021).

Evacuation behaviours vary among social groups for multiple reasons (Kinsey et al., 2019). Some of these relate to decision-making, with demographic factors such as gender shaping social norms and amplifying tendencies to trust or distrust official evacuation messages (Wachinger et al., 2013). More significantly, characteristics such as income, gender, age, and mobility enable or constrain what evacuation options people are aware of, have available and can implement successfully (Deng et al., 2021; Hong et al., 2021). Culturally and linguistically diverse (CALD) populations are less likely to receive or interpret evacuation warnings as intended (Shepherd and van Vuuren, 2014). Groups such as the elderly, individuals with disabilities, and those with limited resources are often unable to evacuate successfully on their own (Kusenbach et al., 2010), and those disadvantaged in multiple ways – e.g. members of CALD communities living with disabilities (Crawford et al., 2021) – are especially impacted (Hong et al., 2021). Availability of transport and transport routes (which can vary significantly throughout a disaster) often strongly influences people's decisions about whether and how to try to evacuate (Lechner and Rouleau, 2019).

'Social capital' (strong social relationships) is the primary determinant of positive disaster outcomes (Aldrich, 2023; Klinenberg, 2015; Melo Zurita et al., 2018), in part because it encourages disaster volunteerism (e.g. Choo and Yoon, 2022). Yet how they influence evacuation actions is complex. On the one hand, community assistance can encourage and enable evacuation, especially among vulnerable groups (e.g. Shoji and Murata, 2021). On the other hand, strong community connections can encourage and enable people to stay in place, especially during the disaster recovery and rebuild phase (e.g. Fraser et al., 2022). Social relationships also shape where people evacuate to and how long they stay away, including whether they return at all (Deng et al., 2021; Fraser et al., 2022). These insights underline how risk is experienced and managed collectively (Kamstra et al., 2021).

Followed by evacuees' behaviour classification introduced by Cova et al. (2011), Shahparvari et al. (2015a), developed a decision support system to identify the number of required vehicles and the optimal evacuation schedule and routes in a predefined period for people who initially decide to shelter in place, however, later decide to leave when it is late.

Hsu and Peeta (2014) discussed a self-evacuation version of this problem, in which a mixed-integer linear programming model was developed to determine the optimal schedule and allocate resources (vehicles) to intersections. An evacuation route planning model considering the uncertainty of demand and capacity was introduced to ensure the reliability of the developed model. Through optimising resource allocation, Shahparvari et al. (2015b) developed a mathematical model to maximise the total number of evacuees. The state-dependent queue roadway link effect in the evacuation network was examined by Moshtagh et al. (2018) using the Stochastic Queue Core (SQC) model. To better understand road maps in disaster-hit areas, Amideo et al. (2019) evaluated shelter locations and evacuation routes.

Based on the road network's capacity and the threat's evolution, Pillac et al. (2015) presented a mathematical model to optimise and mobilise evacuation planning. To advance the prediction of evacuees' behaviour (Anyidoho et al., 2022) compared comprehensively five different models according to their practical utility for future hurricanes. They concluded that the Logistic Regression (LR) is easiest for implementation when only an estimate of the total evacuation rate is necessary. While Dynamic Discrete Choice (DDC) is recommended in case spatial and/or timing predictions are necessary. Additionally, Jin et al. (2021) investigated pedestrian evacuation planning and the location selection of underground emergency shelters for urban regions when a disaster occurs. A round-trip bus evacuation model is developed by Zhao et al. (2020) according to the unfixed routes that optimise the bus scheduling and routing planning, aiming to minimise the total evacuation time and cost. Two algorithms (i.e., the insertion and edge-exchange algorithms) are integrated to obtain the model's sub-optimal solution. Lopez-Carmona (2022) modelled crowd dynamics by introducing a system identification used as a model predictive control for behavioural modification. The aim was to generate distance-keeping instructions to control pedestrian flow optimally. A general framework is presented by Zhang and Zhang (2022) based on multi-trip Electric Bus (EB) models to examine the feasibility of EBs' application during evacuation. The EBs' charging behaviour is also studied along with its economic benefits.

A conceptual framework is presented by Thakur et al. (2022) based on the socio-psychological approach to examine evacuation intention using an online behavioural expectations questionnaire survey in Auckland, New Zealand. Moreover, Chen et al. (2022) collected tsunami evacuation logistics data using a questionnaire survey to analyse the Tsunami arrival time and clearance time (preparation and travel) using probabilistic functions. The structural equation modelling is applied to implement the developed framework.

An emergency planning and scheduling study was conducted later using CG method by Pillac et al. (2016). The same method is used to investigate the multi-trip location routing problem by Ramos et al. (2018). Capelle et al. (2019) adopted a CG scheme for the location and routing problem with pick-up and delivery. The Label Setting Algorithm (LSA) was then applied for the shortest path

problem with pick up & delivery and time window. They concluded that their model and the proposed solution approach could solve large-scale instances in a maximum time of one hour. In contrast, the model without the developed solution approach can only solve small instances. The *B&P* approach is a variant of the *CG* method that has been investigated by Gharaei and Jolai (2019), Goerigk et al. (2013) for the mass evacuation problem to minimise the total clearance time. Goerigk et al. (2014) applied the *B&P* approach to solving location and scheduling problems while achieving a minimum evacuation time. Dayarian et al. (2015) proposed a set partitioning model that uses *B&P* to solve the multi-period evacuation problem. The Branch and Price and Cut (*B&P&C*) algorithms incorporating bucket graph-based label algorithms were used to solve the two-echelon capacitated vehicle routing problem. To investigate heterogeneous green vehicle routing with time limitations, Yu et al. (2019) proposed a novel branching strategy that can significantly reduce the number of branches and computational time.

As disabled and elderly people have limited mobility, they need more supporting items and resources during evacuations (i.e., wheelchairs, walking frames, and medications) (Tomek et al., 2014). An agent-based model capable of simulating evacuation behaviour was developed by Nakanishi et al. (2019) to manage elderly and disabled people in a disaster in the Japanese town of Mannō in the Nakatado district of the Kagawa Prefecture. Chakraborty et al. (2019) studied the effect of flood caused by the Harvey Hurricane on disabled people's evacuation process. They found that people with disabilities with cognitive and ambulatory difficulties were more vulnerable to natural or human-made disasters in the evacuation process.

Luo et al. (2021) analysed the unbalanced transportation network problem regarding fuel consumption. A model based on automobile theory was used to calculate fuel consumption, and experiments on random instances proved its applicability. Southworth (1991) defined traffic loading curves, representing the spread of the evacuated population at a given time. A meta-heuristic method based on the Tabu search is developed by Mathlouthi et al. (2021) for the routing scheduling problem considering multiple time windows. They compared the suggested approach with *B&P* algorithm on small instances. The obtained result was not comparable with *B&P* algorithm in large-scale instances. A real-world case study of floods and landslides in Brazil was addressed by Moreno et al. (2020a) by developing three mathematical models and valid inequalities of the heterogeneous multi-crew routing scheduling problem. Moreno et al. (2020b) developed a hybrid Branch & Benders & Cut (*BBC*) method that combines Simulated Annealing (*SA*) with the *BBC* algorithms for routing scheduling problems. As a result of their investigation, a feasible solution was obtained for 390 instances, while an optimal solution was obtained for 30 instances, which resulted in an 87% reduction in computational time. Purba et al. (2022) applied a new *B&P* meta-heuristic approach to solving the large-scale evacuation planning problem considering refuelling.

This section compares related studies, structurally similar models, and features characterising the evacuation problem. No previous research has provided any solutions for optimising the integration of mass and self-evacuation problems. While structurally similar problems, self-evacuation (Afkham et al., 2022) and mass evacuation (Li et al., 2022), are helpful as a baseline for formulating and solving the evacuation problems, no other published research has focused on the mass evacuation (i.e., evacuation of the elderly and disabled residents) and self-evacuation, simultaneously. This study addresses the evacuation problem, especially for large-scale transportation networks, in two aspects. First, from a modelling standpoint, the earliest arrival time considering the evacuation traffic flow pattern is adopted. Second, to improve the computational efficiency, the novel algorithm *DBNSA* is integrated into the *CG* algorithm as a part of *B&P* algorithm to solve large-scale *MINLP*. A simple formulation is proposed to estimate traffic congestion considering various measures. This leads to the Strategic Evacuation Route Scheduling Problem (*SERSP*), which deals with the above-mentioned challenges.

3. Research aims and contributions

This research aims to answer the following research question: *What is the optimal plan to evacuate the maximum number of residents in the least amount of time as part of a disaster response?* This study models a practicable and executable logistics scheduling system for evacuations in disaster-hit areas. The presented mathematical model provides a practical and effective scheduling approach to maximise the total number of people evacuated, focusing on people with a disability during a rigid time frame whilst tackling congestion along the evacuation paths.

In this study, a novel approach is proposed to address the challenge of evacuating elderly and disabled people during a catastrophic disaster. The paper introduces a DSS that formulates traffic balance for regular and assisted evacuation and considers traffic congestion in the evacuation clearance time by applying contraflow. A Branch and Price algorithm is developed to solve the proposed model in large-size problems. The paper contributes to transportation knowledge by providing a new method for managing evacuation during disasters while improving the resilience of communities through optimal strategic and operational decisions. The traditional methods lack consideration of the specific needs of the elderly and disabled individuals during evacuation.

This study develops a novel model to plan to evacuate residents, including the elderly and disabled people, optimally. A novel heuristic algorithm, the Dominance-Based Neighbourhood Search Algorithm (*DBNSA*), is integrated into the Column Generation (*CG*) algorithm to solve the mixed-integer problem.

4. Model development

According to the arguments presented in Section 2, this paper formulates the *SERSP* to solve a complex *MINLP*, which includes a large number of integer variables and rigid constraints. In an evacuation network, the affected points need to be evacuated by a predefined time, which can limit the number of routes available. The strategic evacuation network can be illustrated by a graph $N = (\mathcal{E}, \mathcal{S}, \mathcal{D}', \mathcal{Z}, \mathcal{Z}')$ where $\mathcal{E}$, $\mathcal{S}$, $\mathcal{D}'$, $\mathcal{Z}$ and $\mathcal{Z}'$ are a set of affected nodes, transit nodes, safe nodes (shelter), a set of edges

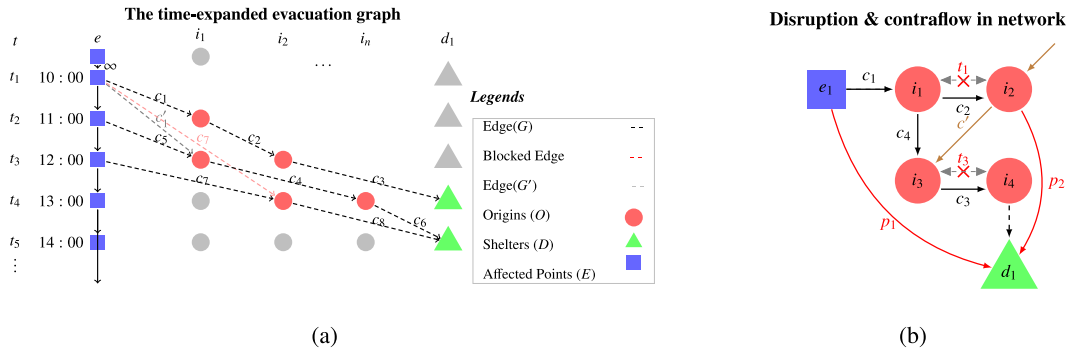


Fig. 1. Fig. 1(a) shows the time-expanded graph for the evacuation scenario; while Fig. 1(b) demonstrates modelling of a contraflow evacuation planning problem scenario.

with current directions, and a set of edges in the opposite directions, respectively. Each evacuated node has several evacuees and a predefined time. Each edge can be characterised by $\bar{b}_{gp}$, u_g and b_g the travel time of the edge, the capacity of the edge, and the time at which edge g becomes unavailable. Fig. 1 provides a schematic of the examined evacuation problem in this study.

Fig. 1(a) demonstrates the progress direction of fire and evacuation flow of affected points over a time window. A plan is scheduled to cover all affected points in each time step. Additionally, 1(b) depicts the implementation of contraflow in those routes that severely affect traffic congestion. For example, routes c_1 and c_2 , two-way roads, are utilised in one direction to use contraflow. The capacity of these routes would double, and traffic congestion would be reduced significantly.

This model aims at getting everyone (e.g., the elderly and disabled people) to a shelter by a specified deadline. However, the nature of a disaster limits the time of the evacuation process (Zhang et al., 2015). Thus, the size of the evacuation problem, network topology, and travel conditions determine the total clearance time (Zhang et al., 2019). The proposed model can determine whether edge g is utilised in contraflow, the departure time of vehicle v to transit the elderly and disabled residents from affected points to shelters, and whether path p is applied in the evacuation process.

4.1. Mathematical formulation

The *SERSP* addressed in this study has the following assumptions:

- Shelters' location and capacities are predetermined.
- The evacuation of elderly and disabled people starts from a sink node (dummy node) and ends in shelters.
- Every affected point has a known number of evacuees.

According to the assumptions as mentioned earlier, the *SERSP* is formulated as the *MINLP* model using a list of sets, indices, parameters, and decision variables that are shown in Table 1.

$$\text{Max } z = \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't} + X'_{po'd't}] \quad (1)$$

Subject to:

$$\sum_{p \in \mathcal{O}'(\mathcal{K})} Z_p = 1 \quad \forall k \in \mathcal{A} \quad (2)$$

$$\sum_{p \in \mathcal{G}_j^-} Z_p - \sum_{p \in \mathcal{G}_j^+} Z_p = 0 \quad \forall i \in \mathcal{S}, \mathcal{S} \in \mathcal{J} \quad (3)$$

$$\sum_{p \in \mathcal{O}'(\mathcal{K})} \sum_{t \in \mathcal{B}'(\mathcal{T})} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't} + X'_{po'd't}] + \bar{X}_i = j_i \quad \forall i \in \mathcal{A} \quad (4)$$

$$\sum_{p \in \mathcal{W}} \sum_{t - \bar{b}_{gp} \in \mathcal{B}'_{\mathcal{G}}} X_{po'd't} \leq u_g \quad \forall g \in \mathcal{G}, t \in \mathcal{T}, o' \in \mathcal{O}', d' \in \mathcal{D}' \quad (5)$$

$$\sum_{p \in \mathcal{W}} \sum_{t - \bar{b}_{gp} \in \mathcal{B}'_{\mathcal{G}}} X_{po'd't} \leq Y_g u_g + (1 - Y') u'_g, \quad \forall g \in \mathcal{G}, t \in \mathcal{T}, o' \in \mathcal{O}', d' \in \mathcal{D}', g' \in \mathcal{G}' \quad (6)$$

$$\sum_{t \in \mathcal{T}} \sum_{d' \in \mathcal{D}'} \sum_{o' \in \mathcal{O}'} X_{po'd't} \leq b'_p Z_p u''_p \quad \forall p \in \mathcal{M} \quad (7)$$

$$Y_g + Y'_g \leq 1 \quad \forall g \in \mathcal{G}, g' \in \mathcal{G}' \quad (8)$$

$$\sum_{o' \in \mathcal{O}'} \sum_{d \in \mathcal{N}} \bar{Y}_{vd o'} \geq 1 \quad \forall v \in \mathcal{V} \quad (9)$$

Table 1

Nomenclature; list of sets, indices, parameters, and decision variables.

Sets			
$\mathcal{A}$	Set of evacuation nodes; $\mathcal{A} = \{1, 2, \dots, A\}$	$\mathcal{B}'_p$	Set of time steps which path p is usable; $\mathcal{B}'_p = \{1, 2, \dots, B'_p\}$
$\mathcal{C}$	Set of edges that can be used in contraflow; $\mathcal{C} = \{1, 2, \dots, C\}$	$\mathcal{C}'$	Set of edges that cannot be used in contraflow; $\mathcal{C}' = \{1, 2, \dots, C'\}$
$\mathcal{D}$	Set of vehicles depots; $\mathcal{D} = \{1, 2, \dots, D\}$	$\mathcal{D}'$	Set of destination (shelters) nodes; $\mathcal{D}' = \{1, 2, \dots, D'\}$
$\mathcal{E}$	Set of affected points; $\mathcal{E} = \{1, 2, \dots, E\}$	$\mathcal{E}$	Set of edges; $\mathcal{E} = \{1, 2, \dots, G\}$
$\mathcal{E}'$	Set of edges going in the opposite direction of edge G ; $\mathcal{E}' = \{1, 2, \dots, G'\}$	$\mathcal{E}^+_i$	Set of incoming edges of node i ; $\mathcal{E}^+_i = \{1, 2, \dots, G\}$
$\mathcal{E}^-_i$	Set of outgoing edges of node i ; $\mathcal{E}^-_i = \{1, 2, \dots, G\}$	$\mathcal{F}$	Set of all nodes; $\mathcal{F} = \{1, 2, \dots, I\}$
$\mathcal{K}$	Set of all evacuated nodes; $\mathcal{K} = \{1, 2, \dots, K\}$	$\mathcal{M}$	Set of all feasible paths between evacuated node k and shelters d' ; $\mathcal{M} = \{1, 2, \dots, M\}$
$\mathcal{N}$	Set of points with passable road; $\mathcal{N} = \{1, 2, \dots, N\}$	$\mathcal{O}'$	Set of origin nodes; $\mathcal{O}' = \{1, 2, \dots, \infty'\}$
$\mathcal{P}$	Set of paths; $\mathcal{P} = \{1, 2, \dots, P\}$	$\mathcal{S}$	Set of transit nodes; $\mathcal{S} = \{1, 2, \dots, S\}$
$\mathcal{T}$	Set of time steps; $\mathcal{T} = \{1, 2, \dots, T\}$	$\mathcal{V}$	Set of vehicles; $\mathcal{V} = \{1, 2, \dots, V\}$
$\mathcal{W}$	Subset of paths that contain edge g ; $\mathcal{W} = \{1, 2, \dots, W\}$		
Indices			
a	Index of evacuation nodes; $a \in \mathcal{A}$	b'_p	Index of time step in which path p is accessible; $b'_p \in \mathcal{B}'_p$
c	Index of edges that can be used in contraflow; $c \in \mathcal{C}$	c'	Index of edges that cannot be used in contraflow; $c' \in \mathcal{C}'$
d	Index of vehicle depots; $d \in \mathcal{D}$	d'	Index of destination nodes (shelters); $d' \in \mathcal{D}'$
e	Index of affected points; $e \in \mathcal{E}$	g	Index of edges; $g \in \mathcal{E}$
g'	Index of edges going in the opposite direction of edge g ; $g' \in \mathcal{E}'$	g^+_i	Index of incoming edges of node i ; $g^+_i \in \mathcal{E}$
g^-_i	Index of outgoing edges of node i ; $g^-_i \in \mathcal{E}$	i	Index of all nodes; $i \in \mathcal{F}$
k	Index of evacuated nodes; $k \in \mathcal{K}$	m	Index of all feasible paths between evacuated node k and shelters d' ; $m \in \mathcal{P}$
n	Index of point with passable roads; $n \in \mathcal{N}$	o'	Index of origin nodes; $o' \in \mathcal{F}$
p	Index of paths; $p \in \mathcal{P}$	s	Index of transit nodes; $s \in \mathcal{S}$
t	Index of time step; $t \in \mathcal{T}$	v	Index of vehicles; $v \in \mathcal{V}$
w	Index of paths that contain edge g ; $w \in \mathcal{P}$		
Parameters			
b_g	Time step at which edge g becomes unavailable	$\bar{b}_{gp}$	Number of time steps required to reach edge g when following path p
$bigM$	Big amount	j_i	Number of evacuees of node i
j'_o	Number of disabled evacuees of origin node o	l_v	Capacity of vehicle v
n'_g	Number of points with passable edge g	u_g	Capacity of edge g
$u'_{g'}$	Capacity of edge g'	u''_p	Capacity of path p
Decision variables			
$X_{po'd't}$	Number of evacuees that starts evacuation on path p at time t from origin node o' to destination node d'	$X'_{po'd't}$	Number of evacuees with disability that starts evacuation on path p at time t from origin node o' to destination node d'
$\bar{X}_i$	Number of non-evacuated evacuees of node i	Y_g	Binary variable; equals 1 if edge g is used in it's normal direction otherwise, 0
$Y'_{g'}$	Binary variable; equals 1 if edge g is not used in its normal direction otherwise, 0	$\bar{Y}_{vd'o'}$	Binary variable; equals 1 if vehicle v departs node d to affected point o' , otherwise 0
Z_p	Binary variable; equals 1 if path p is selected, otherwise, 0	$Z'_{vo'}$	Auxiliary and sequential variable; the number of nodes being visited by vehicle v in the sub-tour elimination constraints
Z''_{vd}	Auxiliary and sequential variable; the number of nodes being visited by vehicle v in sub-tour elimination constraints		

$$\sum_{o' \in \mathcal{N}} \sum_{d \in \mathcal{N}} \bar{Y}_{vd'o'} \geq 1 \quad \forall v \in \mathcal{V} \quad (10)$$

$$\sum_{o' \in \mathcal{O}'} \sum_{d \in \mathcal{N}} \bar{Y}_{vo'd} j_i \leq l_v \quad \forall v \in \mathcal{V}, i \in \mathcal{F} \quad (11)$$

$$Z'_{vo'} - Z''_{vd} + n'_g \bar{Y}_{vo'd} \leq n'_g - 1 \quad \forall v \in \mathcal{V}, o' \in \mathcal{O}', d \in \mathcal{N}, g \in \mathcal{E} \quad (12)$$

$$Z'_{vo'} \leq n'_g \quad \forall v \in \mathcal{V}, o' \in \mathcal{O}', g \in \mathcal{E} \quad (13)$$

$$Z_p \bar{b}_{gp} \geq b_g bigM \quad \forall g \in \mathcal{E}, p \in \mathcal{P} \quad (14)$$

$$\sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} (X_{po'd't} + X'_{po'd't} \bar{Y}_{vo'd}) + \sum_{i \in \mathcal{I}} \bar{X}_i = \sum_{p \in \mathcal{P}} Z_p u''_p + \sum_{g \in \mathcal{G}} Y_g u_g + \sum_{g' \in \mathcal{G}'} (1 - Y'_{g'}) u'_{g'} \quad \forall t \in \mathcal{T} \quad (15)$$

$$X_{po'd't}, X'_{po'd't} \in \mathbb{Z} \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T} \quad (16)$$

$$Z_p \in \mathbb{Z}_2 \quad \forall p \in \mathcal{P} \quad (17)$$

$$Y_g \in \mathbb{Z}_2 \quad \forall g \in \mathcal{G} \quad (18)$$

$$Y'_{g'} \in \mathbb{Z}_2 \quad \forall g' \in \mathcal{G}' \quad (19)$$

$$\bar{Y}_{vo'd} \in \mathbb{Z}_2 \quad \forall v \in \mathcal{V}, o' \in \mathcal{O}', d \in \mathcal{N} \quad (20)$$

$$Z'_{vo'} \in \mathbb{Z} \quad \forall v \in \mathcal{V}, o' \in \mathcal{O}' \quad (21)$$

$$Z''_{vd} \in \mathbb{Z} \quad \forall v \in \mathcal{V}, d \in \mathcal{N} \quad (22)$$

$$\bar{X}_i \in \mathbb{Z} \quad \forall i \in \mathcal{I} \quad (23)$$

The objective function (1) maximises the total number of evacuees to reach safe nodes (shelters) d' as well as the elderly and disabled residents. Constraint (2) ensures that precisely one path p is used by residents of an evacuated node k to reach safe nodes d' (shelters). Constraint (3) ensures continuity of evacuees use path p to evacuate from affected points e to safe nodes d' . Constraint (4) describes that the total number of evacuated and non-evacuated people ($X_{po'd't}, X'_{po'd't}$) should equal the total number of residents (demand) j_i of affected point e . Constraint (5) explains that the evacuee's flow should meet the capacity of each edge u_g . Constraint (6) ensures that the capacity of edges u_g that can be used in contraflow should be met. Constraint (7) indicates that there would not be a flow of evacuees coming from an affected node e as long as path p is selected. Constraint (8) describes that if edge g is selected in contraflow, edge g' cannot be used in contraflow and would not be utilised. Constraint (9) denotes that the start point d of each vehicle (mass evacuation) v is known. Constraint (10) ensures that each vehicle v should return to start to point d after serving the affected points (i.e., delivering the elderly and disabled residents $X'_{po'd't}$ to safe nodes d'). Constraint (11) assures that the capacity of vehicle l_v should be met. Constraint (12)–(14) are related to sub-tour elimination. These constraints are related to the tour of vehicles v which transit disabled and the elderly evacuees $X_{po'd't}$ from affected points e to shelters d' . If vehicle v services a point e , it will not return to the serviced node. Constraint (15) prevents a time window violation. Path p would not be selected at time steps that are not available. Constraint (16) denotes the balanced flow of evacuees ($X_{po'd't}, X'_{po'd't}$) and non-evacuated evacuees ($\bar{X}_i$) in each time step t . Constraints (17), (18) and (23) show domains of integer variables. Constraints (19)–(22) are binary variables.

As the constraint (15) is non-linear (i.e., multiplication of a binary and an integer variable), according to the Eqs. (24)–(27), this constraint is converted to a linear counterpart.

$$\lambda = \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} \sum_{t \in \mathcal{T}} \sum_{v \in \mathcal{V}} \sum_{d \in \mathcal{N}} X'_{po'd't} \bar{Y}_{vo'd} \quad (24)$$

$$\lambda \leq \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} \sum_{t \in \mathcal{T}} X'_{po'd't} \quad (25)$$

$$\lambda \leq \text{Big } M \sum_{v \in \mathcal{V}} \sum_{o' \in \mathcal{O}'} \sum_{d \in \mathcal{N}} \bar{Y}_{vo'd} \quad (26)$$

$$\lambda \geq \left(\sum_{v \in \mathcal{V}} \sum_{o' \in \mathcal{O}'} \sum_{d \in \mathcal{N}} \bar{Y}_{vo'd} - 1 \right) \text{Big } M + \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} \sum_{t \in \mathcal{T}} X'_{po'd't} \quad (27)$$

4.2. Traffic congestion projection

Investigation of traffic congestion during an evacuation mission has been identified as a vital component of strategic evacuation planning, which includes a projection of clearance time (Jha et al., 2004). Due to the lack of congestion anticipation tools and approaches, traffic jams are often disregarded in strategic evacuation scheduling. In addition, the transportation network structure is unsuitable for mass evacuation (Alam and Habib, 2019). Consequently, traffic volume during many incidents has been problematic and uncontrollable. A formulation based on Shahparvari et al. (2016) is developed here to consider traffic congestion and run through its impact on the evacuation process to make better scheduling decisions. Let $\mathcal{K}'$ be the set of routes, $\mathcal{S}$ the set of road sections, and $\mathcal{V}$ the set of vehicle's type. By a rough approximation of traffic jams, there will be a dramatic decrease in clearance time in evacuation using efficient traffic-calming approaches.

$$\lambda_{ijk'} v = \frac{\sum_{l \in \mathcal{S}} \beta_{k'l} v d_{lk'}}{\sum_{l \in \mathcal{S}} d_{lk'}} \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, k' \in \mathcal{K}', v \in \mathcal{V} \quad (28)$$

$$\lambda' = \gamma_{ijk'} s_v (1 + \alpha) + t \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, k' \in \mathcal{K}', v \in \mathcal{V} \quad (29)$$

$$\bar{C} = \frac{\sum_{v \in \mathcal{V}} n_v \lambda'}{\sum_{v \in \mathcal{V}} \sum_{k' \in \mathcal{K}'} \lambda_{ijkv}} \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (30)$$

$$\lambda_{ijk'} v \in \mathbb{Z}^+ \quad \forall i \in \mathcal{I}, j \in \mathcal{J}, k' \in \mathcal{K}', v \in \mathcal{V} \quad (31)$$

$$\lambda' \in \mathbb{R}^+ \quad (32)$$

$$\bar{C} \in \mathbb{Z}_2 \quad (33)$$

Where λ_{ijk}^v depicts the average capacity of route k' for vehicle type v , $\beta_{k'l}$ previously predicted the capacity of section l of route k for vehicle v , $d_{lk'}$ geometric distance of section l of route k' , λ' route resistance function, $\gamma_{ijk'}$ travel time of route k' between affected point i and safe node j , s_v speed of vehicle v , α time inflation per cent of the trip due to road congestion (i.e., inflation factor) (Cairo Traffic Congestion, 2006), t dwell time (dwell time = 2.5min Jordan (2019)), n_v number of vehicle type v , $\bar{C}$ traffic flow percentage. Eq. (28) calculates the capacity of route k' for vehicle v between two points i and j . Route resistance is formulated in Eq. (29). Eq. (30) estimates traffic volume based on route capacity, which means that it calculates the traffic percentage between two nodes i and j based on factors such as route capacity, route resistance, distance, vehicle type, and vehicle speed. It is considered in travel time and used in the mathematical model to evaluate its impacts on scheduling.

Table A.1 provides detailed data about the travel time between affected points and shelters considering non-traffic and traffic situations.

5. Solution approach - branch & price decomposition

As the presented formulation is sophisticated due to a large number of integer and binary variables, it is not efficient to solve it through a commercial optimisation solver due to the need for high demand on computer memory and time. More efficient algorithms have been developed to cope with this complexity more efficiently (Gharaei and Jolai, 2019). As a result, an exact decomposition algorithm *B&P* is introduced, which is applied in this study. The *B&P* algorithm starts with the Dantzig Wolfe (*DW*) decomposition algorithm and forms the Restricted Master Problem (*RMP*). Afterwards, relaxes the model Lp-relaxation Restricted Master Problem (*LRMP*) and solve it by the simplex method (Algorithm 2-Line 2) (i.e., relaxes the assumption that the variables are integer). Then, the pricing problems should be solved to find columns with negative reduced cost (i.e., minimisation problem) (Algorithm 3-Line 4). If a column is found, it should be added to the *LRMP*; otherwise, the integrality of the solutions should be checked (Algorithm 1-Line 12). Suppose the obtained solutions are integers; the *B&P* algorithm will terminate; otherwise, based on the branching strategies. In that case, it will be branched out on the variable that is not an integer and creates two branches (Algorithm 1-Line 15). Each branch is solved using the above-mentioned process to obtain the optimal solution. These steps will repeat until the obtained values for variables are integers.

Algorithm 1: The Branch & Price (*B&P*) Algorithm

```
// STAGE I: Dantzig Wolfe Decomposition Algorithm (Section Appendix B.1)
1 Generate the Restricted Master Problem (RMP) (Section Appendix B.1.1); // STAGE II: LP Relaxation (Section Appendix B.2)
2 Relax the RMP;
3 RMP ← LRRMP;
4 Generate Dual Relaxed Restricted Master Problem (DRRMP) (Section Appendix B.2.1);
// STAGE III: Column Generation Algorithm (Pricing Problems) (Section Appendix B.3);
5 Generate sub-problems (Sections (Appendix B.3.1–Appendix B.3.2));
6 Generate initial column (Algorithm 3); // STAGE IV: Dominance Based Neighborhood Search Algorithm (Section Appendix B.3.3)
7 while Negative Reduced Cost Matrix = {∅} do
8     Find columns with negative reduced costs to add to LRRMP (Algorithm 4);
9     Solve the pricing problems (Algorithm 3)(Sections Appendix B.3.1, Appendix B.3.2);
10    if there is any column with negative reduces cost then
11        | Add the associated columns to the LRMP;
12    else if all the variables are integer then
13        | Report  $X_{pd'd't}, X'_{pd'd't}, \bar{X}_i$ ;
14    else
15        | Branching Strategies (Section Appendix B.3.4);
```

According to the Algorithm 1, in Stage I Appendix B, the *DW* decomposition algorithm 2 is adopted as one of the most common approaches in decomposing integer programming models to solve large-scale problems (Puchinger et al., 2011; Tonbari and Ahmed, 2019) by pertaining a tighter bound (Vanderbeck, 2000). The problem is then formulated as the *RMP* (Algorithm 1 - Line 1). In the *RMP*, each possible point is generated as a convex combination of some extreme points. In Stage II (Appendix B.1.1), some integer constraints could form polytope (Raidl and Puchinger, 2008), which makes the solution space non-convex. The Lp-relaxation method is then utilised as a common way to make the feasible space convex (Algorithm 2-Line 2). The dual formulation of the *LRMP* problem is then formulated B.2.1. In this form, the optimal solution is found at the extreme point of the convex hull of all feasible integer points (Dey et al., 2013).

Stage III (Appendix B.3) is incorporated with the *CG* algorithm 3 (sub-problem formulation). This approach solves n decomposed sub-problems, iteratively (Algorithm 3-Line 4). Variables associated with the most negative reduced cost (i.e., in minimisation) are added to the *LRPM* (Algorithm 3-Line 6). The optimal solution is found if there is no column with negative reduced cost to add to the *RPM*.

Table 2

Nomenclature; list of sets, indices, parameters, and decision variables for robust model.

Sets			
S'	Set of scenarios		
Parameters			
$c_{d',s'}^1$	Cost of accommodation in shelter d' under scenario s' (establishment of shelter d')	$p'_{s'}$	Probability of realisation of scenario s'
r'	Weight placed on solution variance	$r_{d',s'}^1$	Capacity of shelter d' under scenario s'
u'	Weight assigned to model infeasibility penalty	$j_{is'}$	Number of evacuees of node i under scenario s'
Decision variables			
$Y_{d',s'}^1$	Binary variable takes 1 if shelter d' is selected under scenario s' , otherwise 0	$\lambda'_{s'}$	Infeasibility of the model under scenario s'
$\theta'_{s'}$	Variable to linearise the function		

Stage IV (Appendix B.3.3) is associated with the developed DBNSA 4 to integrate into the CG algorithm to improve the efficacy of finding columns with negative reduced costs (i.e., most positive costs in maximisation problem). This algorithm starts by creating a lower bound. In each iteration, the DBNSA finds neighbours of the initial solution by swap fabrication operator (Algorithm Appendix B.3.3-Line 2) and compares these possible solutions to find the column with the most negative reduced cost (Algorithm Appendix B.3.3-Lines 3–8). The solution is discarded when the objective function value of each candidate solution generated by the swap operator is not better than the bound (Algorithm Appendix B.3.3-Lines 3,4).

In the Branching Strategies' Stage (Appendix B.3.4), if the obtained solutions of Stage VI are integer, the algorithm finishes by returning them as the optimal solution (Algorithm 1-Lines 12,13). The algorithm applies the branching strategies to find integer solutions; otherwise, (Algorithm 1-Line 15) terminates. More detailed explanations of the proposed accelerated B&P are provided in Appendix B.

6. Robustness checking

In this study, the model variation test as the most frequently used test in the literature, (Chen et al., 2007; Neumayer and Plümper, 2017; Choi et al., 2019; Jackson et al., 2019) is applied to evaluate the robustness of the proposed model. The main steps of this method are as follows (Neumayer and Plümper, 2017):

– Step 1: Baseline model generation

The baseline model is generated in Section 4

– Step 2: Baseline model assumption identification, relaxing or replacing

The main assumptions of the proposed model are given in the section mathematical formulation (Section 4.1). This study's first and third assumptions are relaxed and investigated in detail with the baseline model in this test. The relaxation of the second assumption will cause the sub-tour elimination constraints of the model to be violated.

– Step 3: Developing baseline model according to the new/replaced assumption:

The robust mathematical model based on the relaxation of assumptions can be reformulated as follows: The list of sets, indices, parameters, and decision variables is shown in Table 2.

6.1. Model reformulation with relation of the first assumption relaxation:

$$\begin{aligned}
 \text{Min} \quad & \sum_{s' \in S'} p'_{s'} \sum_{d' \in \mathcal{D}'} Y_{d's'}^1 c_{d's'}^1 + \sum_{s' \in S'} p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] \right] + \\
 & r' \sum_{s' \in S'} p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] - \right. \\
 & \left. \sum_{s' \in S'} p'_{s'} \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] + 2\theta'_{s'} \right] + u' \sum_{s' \in S'} p'_{s'} \lambda'_{s'}
 \end{aligned} \quad (34)$$

Subject to:

$$\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] - p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] \right] + \theta'_{s'} \geq 0 \quad \forall s' \in S' \quad (35)$$

$$\sum_{p \in \mathcal{O}'(\mathcal{K})} \sum_{t \in \mathcal{B}'(\mathcal{P})} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd'ts'} + X'_{po'd'ts'}] + \bar{X}_i + \lambda'_{s'} = j_{is'} \quad \forall i \in \mathcal{A}, s' \in S' \quad (36)$$

$$\sum_{p \in \mathcal{W}} \sum_{t \in \mathcal{B}_{gp}} X_{po'd'ts'} + \lambda'_{s'} = u_g \quad \forall g \in \mathcal{C}, t \in \mathcal{T}, o' \in \mathcal{O}', d' \in \mathcal{D}', s' \in S' \quad (37)$$

$$\sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} X_{po'd't's'} + \lambda'_{s'} = Y_g u_g + (1 - Y') u'_{g'}, \quad \forall g \in \mathcal{G}, t \in \mathcal{T}, o' \in \mathcal{O}', d' \in \mathcal{D}', g' \in \mathcal{G}', s' \in \mathcal{S}' \quad (38)$$

$$\sum_{i \in \mathcal{I}} \sum_{d' \in \mathcal{D}'} \sum_{o' \in \mathcal{O}'} X_{po'd't's'} + \lambda'_{s'} = b'_p Z_p u''_p \quad \forall p \in \mathcal{M}, s' \in \mathcal{S}' \quad (39)$$

$$\sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} (X_{po'd't's'} + X'_{po'd't's'} \bar{Y}_{vo'd'}) + \sum_{i \in \mathcal{I}} \bar{X}_i + \lambda'_{s'} = \sum_{p \in \mathcal{P}} Z_p u''_p + \sum_{g \in \mathcal{G}} Y_g u_g + \sum_{g' \in \mathcal{G}'} (1 - Y'_{g'}) u'_{g'}, \quad \forall t \in \mathcal{T}, s' \in \mathcal{S}' \quad (40)$$

$$\sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} (X_{po'd't's'} + X'_{po'd't's'} \bar{Y}_{vo'd'}) + \lambda'_{s'} = r^1_{d's'}, \quad \forall d' \in \mathcal{D}', s' \in \mathcal{S}' \quad (41)$$

$$X_{po'd't's'}, X'_{po'd't's'} \in \mathbb{Z} \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T}, s' \in \mathcal{S}' \quad (42)$$

Eqs. (2)–(3), (8)–(14), (17)–(23).

The robust objective function (34) includes three main parts. The first part minimises expected cost (i.e., maximises the total number of evacuees plus the total number of evacuees that, include elderly people and disabled people). The second term minimises the variance of the total number of evacuees, representing the robustness of the model regarding optimality. Finally, the third term shows the robustness from a feasibility perspective, which means it measures model feasibility by considering a weighted penalty. Concerning equations, (2)–(3), (8)–(14), (17)–(23), no change in the mathematical expression is made compared to the deterministic model. Constraint (35) is added to the model to convert the non-linear objective function to a linear model. To ensure that the penalty is applied in case of unmet demand (i.e., the number of evacuees of each affected point), Eq. (36) is added to the model. Constraint (37) and (38) explain that the evacuation flow should meet the capacity of edges where the contraflow is employed; otherwise, penalty cost will be added to the model. Eq. (39) shows the dependency of evacuees flow to selecting path p . Otherwise, the penalty term will be considered for the model. Furthermore, constraint (40) expresses time window limitations that the model will consider penalty cost if the time window is violated. Finally, the limitations of shelters' capacity are defined by Eq. (41) that $\lambda'_{s'}$ will take a value if the model violates this constraint. Constraint (42) demonstrates integer variables. It should be noted that the difference between Eqs. (34) and (43) is the term related to locating shelters in the first robust model.

6.2. Model formulation with the relation of the third assumption relaxation:

$$\begin{aligned} \text{Min } z = & \sum_{s' \in \mathcal{S}'} p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't's'} + X'_{po'd't's'}] \right] \\ & + r' \sum_{s' \in \mathcal{S}'} p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't's'} + X'_{po'd't's'}] - \right. \\ & \left. \sum_{s' \in \mathcal{S}'} p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't's'} + X'_{po'd't's'}] + 2\theta'_{s'} \right] + u' \sum_{s' \in \mathcal{S}'} p'_{s'} \lambda'_{s'} \right] \end{aligned} \quad (43)$$

Subject to:

$$\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't's'} + X'_{po'd't's'}] - p'_{s'} \left[\sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} [X_{po'd't's'} + X'_{po'd't's'}] \right] + \theta'_{s'} \geq 0 \quad \forall s' \in \mathcal{S}' \quad (44)$$

Eqs. (2)–(3), (8)–(14), (17)–(23), (36)–(40), (42).

The explanation for the Eqs. (2)–(3), (8)–(14), (17)–(23), (36)–(40), (42) are the same as the first robust model. Similar to the first robust model, the objective function of the robust model (43) minimises the total expected cost, variance, and the total penalty regarding the feasibility of the model. Furthermore, Eq. (44) is added to the model to convert the non-linear part of the robust model to the linear one.

– Step 4: Comparing the estimated effects of generated models:

In the comparison, it has been assumed that the optimal solution can be robust if it remains close to optimum and almost feasible in all scenarios (Mulvey et al., 1995; Yu and Li, 2000). For this purpose, three groups of instances, small, medium, and large, are generated (Table 3). Four possibilities for evacuees of affected points as the demand of each affected point (i.e., relaxation of the first assumption) and location of shelters and their capacities (i.e., relaxation of the third assumption) are considered based on the study performed by Pan and Nagi (2010). The uncertain parameters and their corresponding probabilities of realisation of their scenarios are generated uniformly. Briefly, robust models' parameter setting is as follows: $p'_1 = 0.2$, $p'_2 = 0.3$, $p'_3 = 0.3$, $p'_4 = 0.2$, $r' = (0-800)$, $u' = 500$ (Pan and Nagi, 2010). Leveraging robust model parameters from a different study can expedite the development of robust models, especially when data or resources are unavailable. The approach has been widely used in the literature, provided that the underlying system being modelled is structurally similar to the one in the previous study (Ben-Tal and Nemirovski, 1998;

Table 3
Robustness checking.

No.	Instance size					Baseline (Deterministic)		Assumption 1		Assumption 3	
	$\mathcal{E} \times \mathcal{S} \times \mathcal{D}'$	$\mathcal{T}$	$\mathcal{P}$	$\mathcal{C}$	$\mathcal{C}'$	O.F.V	C.T(s)	O.F.V	C.T(s)	O.F.V	C.T(s)
1	$8 \times 16 \times 4$	20	20	12	12	3218.4339	13	16 242.2976	97	15 168.9846	77
2	$16 \times 32 \times 8$	18	18	10	10	3812.4469	18	16 725.1606	389	16 363.7014	89
3	$32 \times 32 \times 10$	20	20	12	12	6350.8337	78	18 118.9024	168	14 889.1479	745
4	$160 \times 8 \times 8$	10	10	10	10	25 644.4761	106	20 335.6652	528	19 015.7892	531
5	$200 \times 8 \times 8$	10	10	10	10	31 591.4267	144	20 867.5619	1448	19 662.7689	1294
6	$300 \times 8 \times 8$	8	8	10	10	46 474.2130	179	24 130.7978	1284	22 849.0351	1336
7	$400 \times 8 \times 8$	8	8	9	9	61 842.2646	495	24 290.9352	1230	23 103.4666	1154
8	$450 \times 8 \times 8$	8	8	8	8	69 598.4413	1004	23 220.8694	1742	22 032.9739	1457

Table 4
Random variables and their distribution.

Parameters	Distribution	Parameters	Distribution
u_g	$\sim \mathcal{N}(100, 200)$	$u'_{g'}$	$\sim \mathcal{U}(100, 200)$
u'_p	$\sim \mathcal{N}(200, 300)$	l_v	$\sim \mathcal{U}(3, 6)$
b_g	$\sim \mathcal{N}(1, 40)$	$\tilde{b}_{gp}$	$\sim \mathcal{U}(1, 4)$
j_i	$\sim \mathcal{U}(100, 200)$	j_o	$\sim \mathcal{U}(1, 15)$
n'_v	$\sim \mathcal{N}(1, 4)$	n_g	$\sim \mathcal{N}(1, 280)$

Delage and Ye, 2010). Previous studies have shown that this approach can lead to more accurate and reliable results, and conducting sensitivity analysis can provide more insight into the appropriateness of using such parameters (Delage and Ye, 2010).

The comparison results of three developed models regarding objective function value and computational time are summarised in Table 3. The objective function of the robust models represents the worst-case scenario under uncertainty. In contrast, the baseline objective function (deterministic) does not consider uncertainty and disturbance in different parameters of the model. According to Table 3, by increasing the sample's size, the robust models result in more robust solutions than the baseline model. The computational time of the baseline model is much lower than the robust models as it is not incorporated with the additional index of s' as scenarios, meaning that it has less complexity than the robust models. By setting u' to a large amount, the model must meet the uncertain parameters (i.e., all evacuees of all affected points should be evacuated under all scenarios). Otherwise, the model considers a considerable penalty cost.

Generally, the r' provides a trade-off between the model's robustness and the solution's robustness. Moreover, the r' controls the solution variance. For example, setting r' indicates the decision-makers focus on the expected number of evacuees and ignore the evacuees' variability. Therefore, the expected number of evacuees will proliferate by increasing r' will proliferate the expected number of evacuees.

The application of robust optimisation on the evacuation problem underlines the potential of both strategies to optimise evacuation scheduling processes. The robust strategy explicitly considers noise variables and optimises probability distributions of objective functions and constraints to achieve a robust and reliable solution. Generally, for all instances, the models can produce robust solutions confirming effectiveness in different uncertain situations and perturbations.

The robustness evaluation is crucial because decision-makers rely on accurate and reliable models to inform their policies and actions during disasters. Inaccurate or biased model results can lead to sub-optimal decisions and actions that severely affect the affected population. By conducting a robustness evaluation, decision-makers can have confidence in the model's results and use them to develop effective and efficient evacuation plans. The robustness evaluation can help decision-makers identify the proposed model's limitations and areas that require further improvement. This information can guide decision-makers in refining the model and developing more robust evacuation plans to withstand various disaster scenarios.

7. Numerical experiments

In this section, three sets of samples, large, medium, and trim, are generated based on random data to prove the capability of the model and solution approach in solving various samples with different sizes. Real data is incorporated with the case study to evaluate the efficiency and effectiveness of the proposed model and solution approach in dealing with real-world emergencies. The number of affected points $\mathcal{E}$ for the three sizes are considered 4–40, 40–160, and 250–280, respectively. Three, five, seven, and ten shelters $\mathcal{D}'$ are facilitated for residents. Mission times are projected at 6, 10, 20, 30, and 40-time steps, respectively. Eight, 16, 20, and 40 transit nodes $\mathcal{S}$ are utilised between affected points and shelters. Five, 10, and 15 vehicles v are prepared to transit the elderly and people with disabilities by a capacity of three, five, and seven persons, respectively. Parameter setting is briefed in Table 4.

Table 5 verifies how the presented model provides a strategic and optimal schedule for the evacuation process in a varying range of regions. Overall, the proposed model solves small samples quickly and increases the network's size; the computational time grows steadily.

Table 5
Numerical experiments results.

No.	Instance size					Non-traffic			Traffic		
	$\mathcal{E} \times \mathcal{S} \times \mathcal{D}'$	$\mathcal{T}$	$\mathcal{P}$	$\mathcal{C}$	$\mathcal{C}'$	O.F.V	C.T(s)	%	O.F.V	C.T(s)	%
1	$4 \times 8 \times 3$	6	6	3	3	1119	10	100%	1000	10	89%
2	$7 \times 8 \times 3$	6	6	3	3	1869	12	100%	1750	12	93%
3	$7 \times 16 \times 3$	6	6	3	3	1869	12	100%	1750	13	93%
4	$7 \times 16 \times 5$	6	6	3	3	1869	14	100%	1750	17	93%
5		10	6	3	3	1869	15	100%	1869	19	100%
6		10	10	3	3	1869	26	100%	1869	31	100%
7		10	10	6	3	1869	31	100%	1869	37	100%
8		10	10	6	6	1869	38	100%	1869	42	100%
9	$20 \times 16 \times 5$	10	10	6	3	5099	252	98%	5000	260	98%
10	$40 \times 20 \times 7$	10	10	6	6	9428	607	90%	9275	623	88%
11		20	20	12	6	9428	1022	90%	9275	1029	88%
12	$160 \times 40 \times 10$	30	30	15	6	101 664	2280	93%	99 813	2301	89%
13		40	30	15	6	102 688	3009	100%	101 429	3067	98%
14	$200 \times 40 \times 10$	40	30	15	6	112 688	3900	100%	100 216	3936	97%
15	$250 \times 40 \times 10$	40	30	15	6	98 620	7134	76%	97 019	7186	74%
16		50	30	15	6	129 560	9116	100%	127 791	9159	98%
17	$280 \times 40 \times 10$	40	30	15	6	136 090	14 412	94%	133 740	14 702	92%

Table 6
Experimental results on the comparison between CPLEX, B&P, Accelerated B&P and the two state of the art B&P in the literature.

No.	Instance size						CPLEX		B&P		Accelerated B&P		B&P (Reihaneh and Ghoniem, 2019)		B&P (Václavík et al., 2018)	
	$\mathcal{E} \times \mathcal{S} \times \mathcal{D}'$	$\mathcal{T}$	$\mathcal{P}$	$\mathcal{C}$	$\mathcal{C}'$	O.F.V	C.T(s)	O.F.V	C.T(s)	O.F.V	C.T(s)	O.F.V	C.T(s)	O.F.V	C.T(s)	
1	$8 \times 16 \times 4$	20	20	12	12	320 000	4	320 000	4	320 000	3	343 245	35	320 000	28	
2	$16 \times 32 \times 8$	18	18	10	10	256 000	106	256 000	158	256 000	152	289 658	109	256 000	81	
3	$32 \times 32 \times 10$	20	20	12	12	348 012	515	348 012	485	348 012	349	363 131	443	348 012	388	
4	$160 \times 8 \times 8$	10	10	10	10	195 070	531	195 070	511	195 070	396	155 876	503	195 070	403	
5	$200 \times 8 \times 8$	10	10	10	10	159 633	679	159 633	630	159 633	593	93 248	759	159 633	577	
6	$300 \times 8 \times 8$	8	8	10	10	138 504	1631	138 504	1231	138 504	1220	139 971	1592	138 504	1227	
7	$400 \times 8 \times 8$	10	10	10	10	–	–	140 792	3000	140 792	2920	137 258	3045	140 792	2613	
8	$450 \times 8 \times 8$	12	12	10	10	–	–	158 176	3632	158 176	3650	184 940	4021	158 176	3902	

The more the affected points increase, the more evacuees should be evacuated based on the available routes, the capacity of routes, available time steps, and the number of edges that can be used in contraflow and their capacities. By increasing the number of transit nodes (intermediate points) and safe nodes (shelters), there is no change in the number of evacuated people. At the same time, there is only a slight rise in computational time. For instance, in samples 3 and 4, it can be seen that there is no change in the number of evacuees by doubling transit nodes and increasing shelters. At the same time, any change in transit or safe nodes affects the evacuation schedule significantly, which means that by increasing the shelters and transit nodes, the developed model tries to provide a more balanced schedule for evacuees. There is a substantial increase in evacuees where the time step increases. For instance, the first-time step's spike is accompanied by a 35% growth in the evacuated population.

Also, paths play the same role as time steps, meaning that traffic congestion will be controlled by increasing the number of available paths. It will reduce, resulting in a proliferation in the number of evacuees. Turning to other factors, such as edges in normal or opposite directions, it can be seen that increasing the number of affected points increases the number of evacuees, and dropping the number of edges that cannot be used in contraflow curbs the number of evacuees. Thereby, contraflow provides a significant rise in the capacity of the respective route; hence, when the number of such edges grows, there will be more capacity for more evacuees to reach shelters sooner without being stuck amid traffic. A more significant rise is predicted when time, routes, and capacities increase sufficiently based on the population. Although the computational time depends more on the time step and paths, the growth of computational time by multiplication of edges that can be used in contraflow is not negligible.

Various instances are generated to investigate and compare the performance of CPLEX, base B&P, the accelerated B&P algorithm, B&P algorithms adopted by Ahmadi-Javid et al. (2018), Reihaneh and Ghoniem (2019) and Václavík et al. (2018) (Table 6). The accelerated version of B&P algorithm in this study and the B&P algorithm adopted by Václavík et al. (2018) are exact-based algorithms that ensure the optimality of the outputs. However, the B&P algorithm adopted by Ahmadi-Javid et al. (2018), Reihaneh and Ghoniem (2019) may not guarantee optimality as its core algorithm for the pricing problem is heuristic-based.

According to Table 6, it can be concluded that the CPLEX solver fails to solve the large-scale instances. The accelerated B&P algorithm outperforms the base version of B&P algorithm in terms of computational time by solving on average 14.26% faster, obtaining the same objective function values. On average, the developed B&P algorithm provides 3.53% better solutions and 12.03% less computational time than the developed B&P algorithm by Reihaneh and Ghoniem (2019).

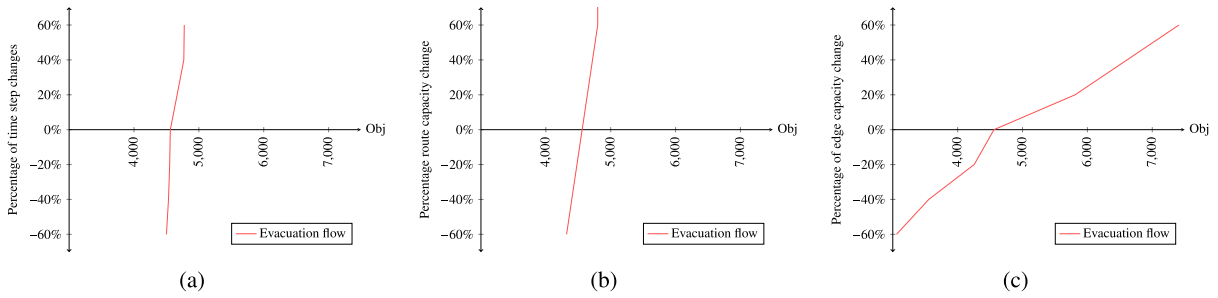


Fig. 2. Alternation of evacuated population based on three key parameters (time steps, the capacity of edges and routes).

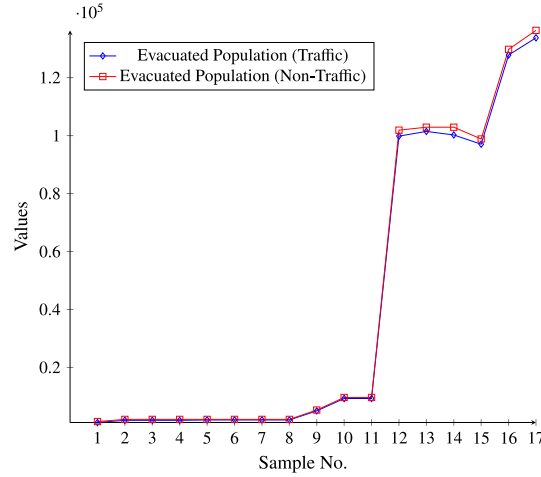


Fig. 3. Analysis of traffic and non-traffic evacuation flow.

The quality of solutions provided by the developed *B&P* by Václavík et al. (2018) is the same as the obtained solution by the accelerated *B&P* algorithm in this study. While, in terms of computational time, which is crucial in emergency-related situations, the accelerated *B&P* algorithm in this study requires approximately 11% less of the computational time on average than the developed *B&P* by Václavík et al. (2018) to obtain the same optimal objective function value (O.F.V). As it can be seen in Table 6, the accelerated *B&P* algorithm in this study requires less computational time in five instances out of all eight instances (e.g., instances 1, 3, 4, 6, 8) that proves the outstanding performance of the developed algorithm in this study.

Fig. 2 illustrates the total number of evacuees according to the various combinations of the time step, route capacity, and edge capacity.

The total number of evacuees' downward and upward trends have the highest fluctuation when the capacity of edge u_g increases or decreases. The more capacity of the edges, the more people are evacuated. For instance, when u_g declines by 20%, the number of evacuated evacuees decreases by approximately 7%. In contrast, reducing the number of time steps by 20% leads to a negligible decrease in the number of evacuees (approximately 1%). Although the capacity of routes and time steps play a significant role in the evacuation process, the crisis management team should devote most of their attention to increasing the capacity of edges, which has the vital role of reducing congestion and total clearance time, potentially leading to a smaller number of fatalities.

Moreover, edge's capacity facilitates transportation for evacuees and relief forces to evacuate quickly with the least fatalities. Route capacity is another significant factor in this problem. The number of evacuees exceeds 4,755 from 4,560 when the capacity of the routes increases by 6%. A typical pattern for evacuating the population for all three factors is that the number of evacuees rises substantially to a peak, followed by a slight growth before plateauing.

Fig. 3 compares the evacuation flow in two conditions (traffic and non-traffic) for 17 samples.

Fig. 3 indicates that the capacity of routes and time steps have less influence, whereas the capacity of edges considerably influences the evacuation network's performance. Consequently, the evacuated population will not drop as long as there is a continuous rise in the capacity of edges (i.e., contraflow). When two-way edges are employed in contraflow, the capacity of this edge will double, and traffic congestion will decrease, meaning there will be a sharp rise in evacuees (Fig. 3). This is also valid for one-way edges that can be used in contraflow, which balances traffic jams and ensures traffic congestion will split more evenly.

Table 7 below summarises the relationships and implications for decision-makers in emergency evacuation planning based on the given findings. Increasing transit nodes and shelters will not affect the number of evacuees significantly, but it will lead to a

Table 7
Implications for decision makers.

Findings	Implications for decision makers
Increasing the number of transit nodes and shelters do not impact evacuee numbers.	Decision-makers should take into account that increasing the number of transit nodes and shelters can provide a more balanced schedule for evacuees. However, it is important to note that simply increasing the number of transit nodes and shelters is not likely to significantly increase the number of evacuees. Other factors, such as the availability of transportation and the decision-making process of individuals, play a more significant role in determining the number of evacuees.
Paths and time steps play a crucial role in the evacuation process, and increasing the number of paths and time steps can significantly increase the number of evacuees.	Decision-makers should consider the impact of paths and time steps on the evacuation process. Increasing the number of paths and time steps can help control traffic congestion and ultimately increase the number of evacuees. By providing more options for individuals to follow during evacuation, decision-makers can ensure smoother traffic flow and accommodate a larger number of evacuees. However, it is important to be mindful of the potential trade-off between increased paths and time steps and longer computational times required for the evacuation planning process.
Contraflow can significantly increase the capacity of a route, allowing more evacuees to reach shelters sooner without being stuck in traffic. Increasing the number of edges that can be used in contraflow can increase the capacity of a route, which can help reduce congestion and total clearance time, potentially leading to fewer fatalities.	Decision-makers should consider implementing contraflow and increasing the number of edges that can be used in contraflow, as it can increase the capacity of a route, help reduce congestion, and potentially lead to fewer fatalities.
Increasing the capacity of edges can significantly increase the number of evacuees, whereas reducing the number of time steps has a negligible effect on the number of evacuees. Increasing the capacity of edges is critical for reducing congestion and total clearance time and can lead to fewer fatalities.	Decision-makers should consider increasing the capacity of edges to reduce congestion and total clearance time, which can lead to fewer fatalities.
The developed B&P algorithm outperforms other algorithms in terms of computational time, and solution quality	Decision-makers should consider using the developed B&P algorithm as it provides better solutions and takes less computational time compared to other algorithms.

more balanced schedule for evacuees. Changes in time steps significantly impact the evacuation process, with a 35% growth in the evacuated population observed for the first time. Increasing the number of available paths will reduce traffic congestion, leading to an increase in evacuees. The number of evacuees will increase when more edges can be used in contraflow. The CPLEX solver fails to solve large-scale instances, so decision-makers should not rely solely on it. The accelerated B&P algorithm outperforms the base B&P algorithm in computational time while providing the same objective function values. The developed B&P algorithm provides better solutions in less computational time than other B&P algorithms. The crisis management team should focus on increasing the capacity of edges to reduce congestion and total clearance time, potentially leading to fewer fatalities.

8. Results and analysis

8.1. Case study

The 2013–14 Australia's bushfire season commenced at the end of winter 2013 and finished by the autumn of 2014 (Trigg et al., 2015). Various regions of the state of Victoria were affected including Jeeralang Junction (i_1), Jeeralang North (i_2), Jeeralang (i_3), LeRoy (i_4), Koornalla (i_5), Callignee north (i_6), Balook (i_7), Callignee (i_8), Callegnee south (i_9) Gormandale (i_{10}) in the Gippsland being heavily burnt. All computations were tested using CPLEX solver 12.8 on a PC with a Quad-Core (2GHZ) CPU and 8G RAM.

The case study contains ten affected points, three shelters, namely Churchill, Traralgon, and Rosedale, and approximately 2,708 residents (ABS, 2011). The evacuation network setting and parameter setting for the case study are as follows:

Ten transit nodes, 11 routes, 47 edges, a horizon of three hours with a time step of 15 min (*i.e.*, 12-time steps) (Pillac et al., 2016), the capacity of path p is estimated at 200 vehicles, the number of vehicles and their capacity to transit elderly and disabled is considered as eight and three, respectively (NSW RFS, 2020), the time step at which path p becomes unavailable is considered based on VicRoads (2020). Fig. 4 provides a better understanding of the case study, including the location of the affected and safe places 4(a), potential links between affected, intermediate and safe points 4(b).

8.2. Analysis

The evacuation process starts with a journey where 100 people evacuated in the first time step from Le Roy (i_4), passing through Traralgon Creek Road to reach the Churchill shelter (d_1). This trip takes approximately (~52 min), about three time steps. Although the Traralgon shelter (d_2) is closer to Le Roy (i_4) (~43 min) when compared with Churchill (d_1), considering traffic congestion the model assigned this affected point to Churchill shelter (d_1). At the same time, 100 persons evacuated from Callignee (i_8) to

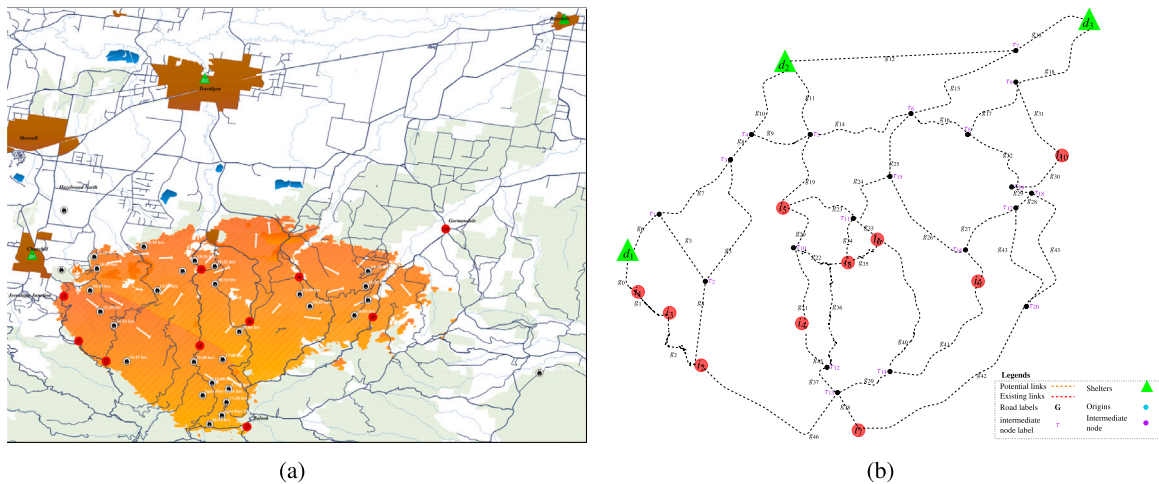


Fig. 4. Case study area; the orange polygons show the area of bushfire propagation and the spread directions (white arrows), the brown polygons show the residential area, the orange fire symbol indicates the origin point of the bushfire, the black fire symbol depicts the location of fire at different time steps. **Fig. 4(a)** Location of affected and safe points; **Fig. 4(b)** Potential links between affected, intermediate and safe points. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Churchill (d_1) by Monash Way (~22 min, almost two-time steps) and 280 persons from Gormandale (i_{10}) to Rosedale (d_3) through Gormandale-Stradbroke Road (~19 min, one-time step) (**Fig. 5(a)**). Approximately 65% of the total population evacuated in the first six-time steps (1:6) and nearly 35% evacuated in the second six time steps (6:12). The travel time for 300 residents of Balook (i_7) in the third time step to the Churchill shelter (d_1) is approximately (~39 min, three-time steps), whereas, the evacuation of the population of Jeeralang North (i_3) (approximately 100 people) to the Traralgon safe node (d_3) takes approximately (~15 min, one time step) (**Fig. 5(b)**). It seems that the Balook's residents (i_7) travel to the Traralgon shelter (d_2) since it is closer than the Churchill shelter (d_1). Due to the unavailability of this route at this time step, the evacuees are assigned to Churchill (d_1) as a safe node. In generating the evacuation plan, the evacuees have been split into different time steps to minimise congestion. For example, 17%, 18%, and 14% of the total evacuee population reach the safe nodes in the first, second and third time steps, respectively. By time step t_7 , the four affected points' evacuees (i.e., i_2 , i_6 , i_9 , i_{10}) reach safe nodes completely (**Fig. 5(g)**).

As traffic is a substantial factor in emergencies, minimising congestion is essential to investigate the case study thoroughly. Contraflow is employed as an efficient solution to maximise the paths' potential capacity. **Fig. 6** provides a better understanding of contraflow and traffic congestion implementation. According to **Fig. 6(a)** in time step three, as the traffic has congested main evacuation routes, edges $g_0, g_1, g_2, g_6 : g_{11}, g_{19} : g_{21}, g_{38}, g_{39}$ and g_{45} are employed in contraflow to disperse the congestion. Almost all of the population at points i_1 , i_2 , and i_3 evacuated at this time step, increasing the traffic in route p_4 (**Fig. 6**). Furthermore, the traffic at points i_4 , i_5 , and i_7 are highly congested. There is light traffic at points i_6 , i_8 , i_9 , and i_{10} . At the next time step, the traffic flow is moved towards points i_6 , i_8 , i_9 . Traffic congestion in p_4 has experienced no changes between time steps t_3 and t_6 . Semi-heavy traffic at points i_1 , i_2 and i_3 at time step t_3 has changed to light traffic at time step six (**Figs. 6(a)** and **6(b)**). If the elderly and disabled residents cannot leave the affected points, it potentially leads to casualties and/or fatalities. The model considers the evacuation of those people to be more compatible with real-world emergencies. Almost 14 disabled people at point i_2 are evacuated in the first time step to shelter d_1 (**Fig. 5(a)**), evacuating 13 people in the next step. The model is scheduled to transfer five, two, and six people with disabilities from i_4 , i_8 and i_{10} to shelters d_1 and d_3 , respectively (**Fig. 5(b)**).

8.3. Heavy traffic congestion scenario

Congestion increases evacuation time, and contraflow is one of the more appropriate solutions to minimise congestion. To further investigate the impact of traffic congestion without using contraflow, a scenario is considered where contraflow is ignored to examine the clearance time and availability of routes. To scrutinise the impacts of congestion precisely, the number of edges that can be used in contraflow is considered zero for the following scenario. As such, there will be an increase in traffic, leading to an increase in clearance time (See **Table A.1**). The developed model provides an optimal strategic plan under heavy traffic conditions without using contraflow.

According to this plan, the total clearance time peaks around 60% to evacuate 2,708 evacuees. This plan can evacuate 66% of the population in 12-time steps. Without any efficient approach to deal with congestion, the number of evacuees decreases significantly to approximately 1,787 people. Residential evacuation is more evenly split between Churchill (d_1) and Traralgon (d_2) (i.e., 45% and 36%, respectively), while only 18% of evacuees are matched by the Rosedale safe node (d_3). The results confirm a correlation between decreasing the number of evacuees and increasing the total clearance time by increasing the traffic volume. Generally, the results show that most affected points were evacuated almost completely, where those points are established in regions with a light

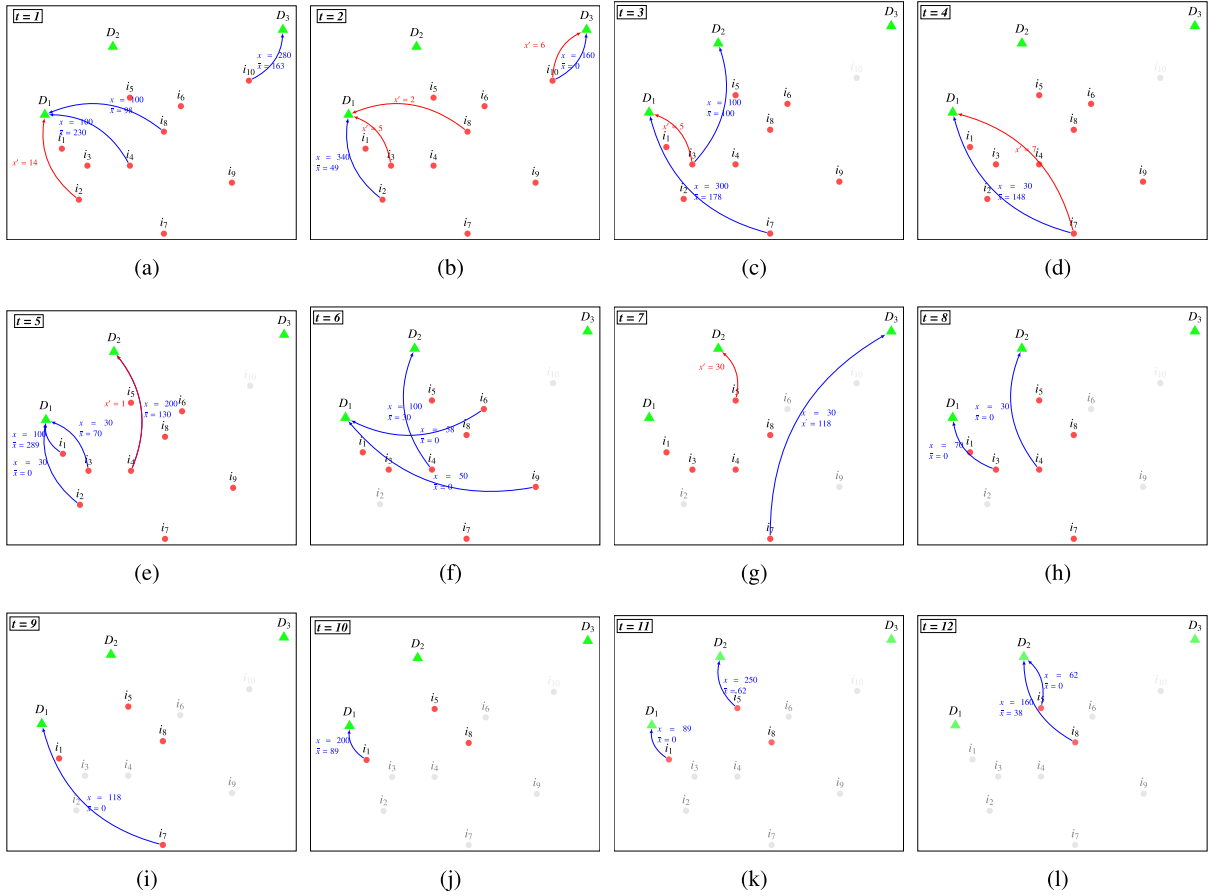


Fig. 5. Total maximum flow for the arterial road; **Fig. 5(a):** Evacuation of the affected points (i_2 , i_4 , i_8 , i_{10}), **5(b):** Evacuation of the affected points (i_2 , i_3 , i_5 , i_8 , i_{10}), **5(c):** Evacuation of the affected points (i_3 and i_7) (population of point i_7 are evacuated completely), **Fig. 5(d):** Evacuation of the affected points (i_7), **Fig. 5(e):** Evacuation of the affected points (i_1 , i_2 , i_3 , i_4) (population of point i_2 are evacuated completely), **Fig. 5(f):** Evacuation of the affected points (i_5 , i_6 , i_9) (population of points i_6 and i_9 are evacuated completely), **Fig. 5(g):** Evacuation of the affected points (i_5 and i_7), **Fig. 5(h):** Evacuation of affected points (i_3 and i_4) (population of points i_3 and i_4 are evacuated completely), **Fig. 5(i):** Evacuation of the affected points (i_7) (population of point i_7 are evacuated completely), **Fig. 5(j):** Evacuation of the affected points (i_1), **Fig. 5(k):** Evacuation of the affected points (i_1 and i_5) (population of points i_1 and i_5 are evacuated completely), **Fig. 5(l):** Evacuation of the affected points (i_8)

traffic flow. For instance, points i_9 and i_{10} are evacuated completely since the traffic sign is light (green). In contrast, only 4% of point i_4 are evacuated via route p_5 , which suffered heavy traffic congestion. Fig. 5 illustrates the evacuation flow at various time steps.

8.4. Sensitivity analysis

To further examine the effects of parameters (i.e., availability of routes at each time step, the number of time steps, the capacity of edges and routes) accurately, a set of scenarios is run to identify the optimal schedule over the predefined period (see Table 8 for scenarios). All parameters are set based on data in Table 4, except for the number of routes (e.g., five main routes are considered).

Fig. 7 compares the various optimal schedules of the evacuation network regarding the number of available routes at different time steps. Regarding Fig. 7, most evacuees are cleared through routes p_4 and p_5 , while routes p_1 and p_2 are barely active during the evacuation process. When the accessibility of route p_4 becomes limited in all 12-time steps (Fig. 7(b)), the model tries to cover the evacuation of the population by relying on p_1 and route p_2 . The capacity of route p_5 appears to be one of the vital routes that evacuate the highest proportion, almost 60% of the population (Fig. 7(b)). The evacuees passing through the third route increase slightly by 6% and a moderate rise in route p_1 , almost 23%. When route p_5 is inaccessible in 12 considered time steps, the evacuation mission commences with 109 evacuees in the first time step being evacuated through routes p_3 and p_4 . The evacuation continues via route p_3 until time step five, where the number of evacuees increases due to evacuation starting at route p_3 . This process will continue by the end of the time step t_5 , which route p_2 will again become unavailable by the end of the time step t_9 and will be available again in the last time step t_{12} .

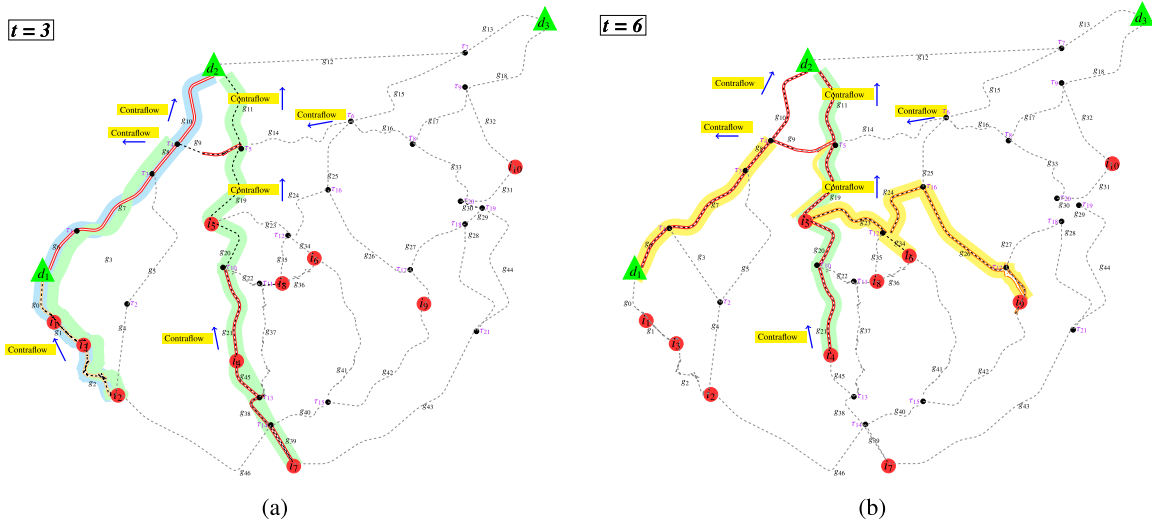


Fig. 6. Traffic flow and implementation of contraflow structure, Heavy traffic: Red lines, Semi-heavy traffic: orange lines, Light traffic: Green lines; **Fig. 6(a):** Traffic flow and the impact of contraflow in the third time step (The flow of evacuees routes 5(c)); **Fig. 6(b):** Traffic current and the impact of contraflow at time step 6 (the flow of evacuees routes 5(f)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 8
Sensitively analysis.

Scenario	$\mathcal{P}$	$\mathcal{T}$	Description
S_1	1:5	1:12	No disruption in all 12 time steps
S_2	1:3,5	1:12	Main route p_4 disrupted in all 12 time steps
S_3	1:4	1:12	Main route p_5 disrupted in all 12 time steps
S_4	1:4	1:6	Main route p_4 disrupted in all 6 time steps
S_5	1:5	1:6	No disruption in all 6 time steps
S_6	2:5	1:6	Main route p_1 disrupted in all 6 time steps
S_7	1,3:5	1:6	Main route p_2 disrupted in all 6 time steps
S_8	1:2,4:5	1:6	Main route p_3 disrupted in all 6 time steps
S_9	1:4	1:6	Main route p_5 disrupted in all 6 time steps

Since response time plays a significant role in emergencies, time steps are decreased from 12 to six to explore the model's capability to generate an evacuation plan within a shorter period. The initial assumption is that all routes are available. Although the first route was not evaluated in previous scenarios, the model provides a balanced plan in route p_1 that can evacuate approximately 663 people during a shorter time frame 7(d). In this situation, the model utilises all resources at total capacity. For instance, route p_2 , which was not active in several time steps, transports 730 evacuees. This shows the flexibility of the presented model in high-risk situations where residents need to be evacuated immediately. The total number of evacuees, in this case, is 2,807 people. There is no fall in the total number of evacuated people through this scenario.

Running through the scenario where the time frame is tighter, routes are now considered blocked due to the severity of the fire. The number of evacuees partially falls by approximately 1% when route p_1 is unavailable. At the same time, the unavailability of the second and third routes affects the network by only 1%. Notably, the first route's capability grows by almost half in this situation. Despite the obtained results for routes p_1 , p_2 , and p_3 , when the time frame includes 12-time steps, these routes have more availability for evacuees. By dropping the time steps to approximately half, the percentage of evacuees through all routes is approximately 99%. Using the developed model, planners can evaluate different shelter placement scenarios to determine which locations have more research roles in supporting and maintaining a balanced distribution of evacuees. This can help ensure that emergency shelters are strategically located to serve the affected population effectively.

9. Managerial insights

As well as the practical benefits of this research as presented above, this research also provides strong implications for strategy and policy development, particularly across a range of stakeholders such as police, emergency authorities, relief forces, traffic management as well as Government - Federal, State/Territory and Local. Specifically, this research provides an evacuation model that will aid stakeholders, primarily police, emergency authorities, relief forces, and traffic management, in developing evacuation strategies that will result in the efficient scheduling of evacuations from impacted areas with minimal interruptions or delays. This

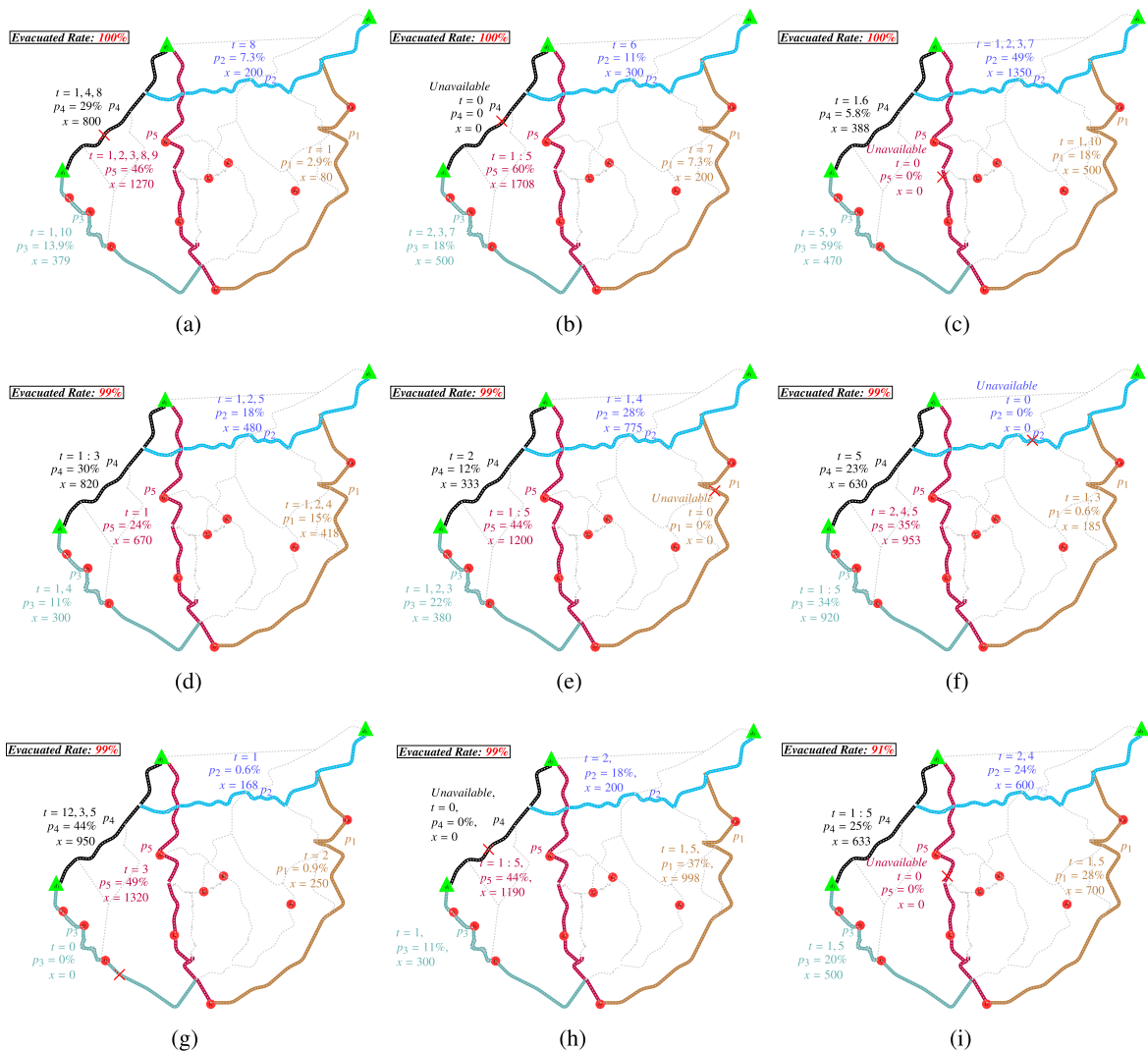


Fig. 7. Evacuation patterns change based on the availability of routes at various time steps. Five main routes of the evacuation network and their associated edges are considered to examine their impacts; **Fig. 7(a):** All routes are available in all 12-time steps; **7(b):** Route p_4 is unavailable in 12-time steps; **7(c):** Route p_5 is unavailable in 5-time steps; **7(d):** All routes are available in 5-time steps; **7(e):** Route p_4 is unavailable in 5-time steps; **7(f):** Route p_5 is unavailable in 5-time steps; **7(g):** Route p_1 is unavailable in 5 time steps; **7(h):** Route p_2 is unavailable in 5-time steps; **7(i):** Route p_3 is unavailable in 5-time steps.

will assist in the rapid scheduling of the evacuations of residents and also factors in road network availability, shelter availability, and speed.

This research will also help with the strategic placement of shelters that will broadly consider the network topology size and response to the evacuation problem, network topology, incorporating the travel conditions and evacuation time-frame whilst considering multiple paths, their availability, and capacity to that shelter. Using the developed model, planners can evaluate different shelter placement scenarios to determine which locations minimise overall evacuation time while maintaining a balanced distribution of evacuees. This can help ensure that emergency shelters are placed strategically to serve the affected population most effectively. Incorporating this model into agencies such as Emergency Management Australia, the Australian Institute for Disaster Resilience, and other related government departments, federal, state/territory, and local would enable authorities to effectively and efficiently guide the withdrawal of residents safely from an emerging threat following a decision and a warning to evacuate.

Section 12.1 of the Australian Royal Commission into National Natural Disaster Arrangements (Binskin et al., 2020, p. 264) highlights that proper planning “for evacuations is an integral part of keeping communities safe from the harm caused by natural disasters. Planning for evacuations should occur early and involve local governments and communities. Evacuation planning should factor in key issues such as seasonal populations, evacuation routes, shelter facilities, jurisdictional boundaries, messaging to communities, and risks to safe evacuation”. Sections 12.20–12.40 of the full royal commission report. As indicated throughout the justification and rationale for this research, the developed model and algorithm address many of these considerations, particularly around the evacuation routes and shelter facilities.

One of the key findings from the Australian Royal Commission into National Natural Disaster Arrangements, and its interim report highlights the difficulties in coordinating and sharing resources between local governments, which sometimes relies on informal understandings (Binskin et al., 2020). The findings from this study support the need for better coordination and resource sharing, and the developed algorithm and model enable such coordination across multiple stakeholders. The developed model also addresses the identified concern for the need for consistent and comparable data and information to be made more widely available, as this can deliver efficiencies, avoid duplication, improve understanding, and facilitate decision-making. Whilst the inclusion of many stakeholders may seem appropriate, it is worth noting the Royal Commission's view that "care is also required to ensure that emergency planning processes remain focused on core objectives, and involve participants who will provide necessary and complementary capabilities, skills and expertise" (Binskin et al., 2020, pp. 256).

In running multiple scenarios across Australia, Emergency Management Australia (EMA) submitted a guidance note to the Royal Commission indicating that they have "undertaken a preliminary scan of jurisdictional emergency disaster management legislation and policy and found consistent themes that establish implied prioritisation criteria" (Department of Home Affairs DHA-Australian Government, 2020, pp. 3). These priorities include "protection of human life is paramount" and "minimising harm and injury to people including those with special needs and authorities".

There is still a need to strengthen the range of options available in the face of fire, including community refuges, bushfire shelters, and evacuation (Teague et al., 2010); ensure that local solutions are tailored and known to communities through local bushfire planning; help municipal councils urgently develop for communities at risk of bushfire local plans that contain contingency options such as evacuation and shelter; document in municipal emergency management plans and other relevant plans facilities where vulnerable people are likely to be situated; and that the State/Territory introduces a comprehensive approach to evacuations so that this option is planned, considered and implemented when it is likely to offer a higher level of protection than other contingency options.

This research will also inform policy, investment, and expectations again at the Federal, state/territory, and local levels. For example, within the state of Victoria, Australia, where the Black Saturday bushfire occurred, the 2009 Royal Commission summary report indicated that the Commission viewed the protection of human life, safety and resilience of communities as the highest priority for bushfire policy and directed its efforts accordingly (Teague et al., 2010).

Finally, to gain the most out of the developed model and algorithm, and as supported by the 2020 Royal Commission (Binskin et al., 2020, pp. 298) finding, there is a need to provide appropriate education by "State and Territory Governments to ensure that the public understands the new system, symbols, terminology and recommended action. The education program must also target people with disabilities, the Indigenous community, and culturally and linguistically diverse Australians". Whilst this is the only reference to people with disabilities, it is still a significant priority in ensuring their safety. Not only does the public understand the new systems, but all stakeholders are aware of the proposed model and algorithm to maintain the priority of protecting all human life.

10. Conclusion

The developed algorithm and model provide an attributable source that enables authorities to identify the most appropriate model to evacuate in response to a disaster. Adopting the developed model would lead to quicker response and reduce the number of casualties and, most importantly, fatalities. The developed model brings together various stakeholders, such as police, emergency authorities, relief forces, traffic management, and affected residents. Additionally, evacuating the elderly and disabled is one of the main concerns for authorities. As such, the presented model enables stakeholders such as emergency authorities and traffic managers to rapidly develop an optimal plan to assign vehicles to transit them to shelters based on route capacity and time.

This study presents a novel model for evacuation planning of regular and assisted evacuees whilst trying to minimise congestion. The development of this model answers the initial research question around the optimal model to evacuate the maximum number of residents in the least amount of time as part of disaster response. Specifically, by evaluating the developed model using data from the 2009 Black Saturday bushfire, we have created a practicable and executable logistics scheduling system for evacuations in disaster-hit areas. The presented model provides a practical scheduling approach to maximise the total number of people evacuated, focusing on the elderly and disabled residents during a tight time frame whilst minimising congestion along the available evacuation paths.

The proposed model adopts the *B&P* approach whose key hallmark is to provide an optimal evacuation plan based on available resources (*i.e.*, available paths, time window, and the capacity of paths and edges) for affected areas using contraflow to tackle congestion. Furthermore, this model focuses on evacuating people with disabilities and the elderly, which has not been explored elsewhere. Estimating traffic congestion enabled the developed model to plan accurately for real-world emergencies.

Findings indicate that the developed model can plan an effective and efficient evacuation schedule for a large sample under various scenarios. Additional simulations indicate that the models' baseline is highly accurate (*e.g.*, particularly in terms of time, which affects the number of evacuees) with a noticeable improvement of 84% of the evacuated population making it to shelters and a 60% improvement in clearance time provided that the local transportation structure has the capability and flexibility of using contraflow and other efficient solutions to reduce traffic congestion. Furthermore, the results show the routes with a high percentage of participation in the evacuation process, which has the most impact on any evacuation plan. The congestion on these routes will increase the total clearance time dramatically. In contrast, traffic jams on routes with a minor role in the evacuation can lead to a negligible rise in the total evacuation time.

Table A.1

Time steps required to reach shelters.

$\mathcal{E} \times \mathcal{S}$	$\mathcal{P}$	$\mathcal{T}$ (Non-Traffic)	mins	C	$\mathcal{T}$ (Traffic)	mins
Jeeralang → Churchill	Linders Road	1	20	0.35	2	27
Jeeralang → Traralgon	C476	1	20	0.31	2	26.2
Jeeralang → Traralgon	C475 Road	1	23	0.35	2	31.05
Jeeralang → Traralgon	Monashway	1	20	0.57	2	36
Jeeralang → Rosedale	Flynns Creek Road	2	36	0.25	3	45
Jeeralang → Rosedale	Princess highway	2	37	0.61	4	59.57
Jeeralang → Rosedale	C476	2	37	0.67	4	61.79
Le Roy → Churchill	Traralgon Creek Road	3	49	0.41	4	69
Le Roy → Churchill	M1	4	57	0.71	6	97.47
Le Roy → Churchill	Jumbak Road	4	56	0.27	3	42.12
Le Roy → Traralgon	Traralgon Creek Road	3	41	0.85	5	75.85
Le Roy → Rosedale	Traralgon Creek Road	4	56	0.3	5	72
Koornalla → Churchill	Traralgon Creek Road	1	21	0.74	2	36.54
Koornalla → Churchill	Monashway	1	23	0.7	3	39.1
Koornalla → Churchill	M1	2	28	0.66	3	46.48
Koornalla → Traralgon	Traralgon Creek Road	1	14	0.9	2	26.6
Koornalla → Rosedale	Traralgon Creek Road	2	28	0.43	3	40.04
Koornalla → Rosedale	Hyland Highway	2	30	0.31	2	40.3
Callignee → Churchill	C476	1	23	1.1	3	48.3
Callignee → Churchill	Monashway	2	30	0.98	4	59.4
Callignee → Churchill	M1	2	30	0.77	4	53.1
Callignee → Traralgon	Callignee S Road	1	16	0.19	1	21.04
Callignee → Rosedale	Flynns Creek Road	2	29	0.28	2	35.84
Callignee → Rosedale	Hylandway	2	31	0.19	2	24.89
Callignee North → Churchill	Callignee S Road	1	24	0.23	2	29.52
Callignee North → Churchill	Monashway	2	26	0.56	3	40.56
Callignee North → Churchill	M1	2	31	0.81	4	56.11
Callignee North → Traralgon	Callignee S Road	1	16	0.98	2	31.68
Callignee North → Rosedale	Hylandway	2	32	0.37	3	43.84
Callignee North → Rosedale	Flynns Road	2	30	0.22	2	36.6
Callignee South → Churchill	Collignee S Road	3	38	0.97	5	73.86
Callignee South → Churchill	M1	3	45	1.3	7	103.5
Callignee South → Traralgon	Collignee S Road	2	31	0.53	3	47.43
Callignee South → Rosedale	Collignee S Road	3	44	0.21	4	52.8
Callignee South → Rosedale	Flynns Road	3	46	0.26	4	57.96
Callignee South → Rosedale	Grand Ridge Road	4	59	0.23	5	72.57

In terms of future research, the limitations of this research include further exploring and incorporating additional factors from the literature that influence the decision-making process during evacuations. Further considerations for animals, livestock, and endangered species and their evacuation could also be included in the next iteration of the proposed model. Further, investigating other types of disasters and disruptions, such as incidents and road closures, would benefit from an adapted version of the developed model. The potential to co-optimize paths, schedules, shelter locations, and other intermediate locations for elderly and disability aggregators, allowing for concurrent planning or simulation, can be further investigated (Siam et al., 2022; Purba et al., 2022). Finally, other disasters, such as floods, which have received little attention thus far, could also be simulated.

CRedit authorship contribution statement

Shahrooz Shahparvari: Conceptualization, Data curation, Investigation, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mahsa Mohammadi:** Formal analysis, Methodology. **Konrad Peszynski:** Writing – review & editing. **Lauren Rickards:** Writing – review & editing.

Data availability

Data will be made available on request.

Appendix A. Data

See Table A.1.

Appendix B. Solution approach- supplementary materials

B.1. STAGE I: Dantzig Wolfe decomposition algorithm

The Dantzig Wolfe Decomposition (*DWD*) algorithm is capable of solving large-scale problems (Tonbari and Ahmed, 2019). In order to solve an integer programming model, the *DWD* is an ideal way that decompose an integer model into a master problem and several sub-problems (Puchinger et al., 2011). The associated master and sub-problems relate to optimising the flow of evacuees and providing a set of feasible paths respectively to provide optimal scheduling. Since *DWD* is implemented on an integer programming model, it is a specific reformulation of the problem to improve a tighter bound (Vanderbeck, 2000).

B.1.1. Restricted Master Problem (RMP)

The *DWD* formulation of the problem is generated by the *RMP*. It should be noted that, since, in *DWD* each feasible point is formulated as a convex combination of extreme points η and for each constraint $A_i X_i \leq d_i$, then $\sum_{j=1}^{k_i} \eta_{ij} = 1$. The formulation of *DWD* is given as follow:

$$\text{Min } RMP = \sum_{j_1=1}^{k_1} \eta_{j_1}^1 + \sum_{j_2=1}^{k_2} \eta_{j_2}^2 \quad (\text{B.1})$$

Subject to:

$$\sum_{j_m=1}^{k_m} \eta_{j_m}^m \leq 1 \quad \forall m \in \mathcal{M}; (\mathcal{M} = \{1, \dots, 8\}) \quad (\text{B.2})$$

$$\sum_{j_1=1}^{k_1} \eta_{j_1}^1 + \sum_{j_4=1}^{k_4} \eta_{j_4}^4 \leq \sum_{j_8=1}^{k_8} \eta_{j_8}^8 + \sum_p u_p''' \sum_{j_3=1}^{k_3} \eta_{j_3}^3 + \sum_{g \in \mathcal{G}} u_g \sum_{j_5=1}^{k_5} \eta_{j_5}^5 + \sum_{g' \in \mathcal{G}'} u_{g'}' \sum_{j_6=1}^{k_6} \eta_{j_6}^6 \quad (\text{B.3})$$

$$\eta_{j_m}^m \in \mathbb{Z} \quad m \in \mathcal{M}; (\mathcal{M} = \{1, \dots, 8\}), j \in \{1, \dots, N\} \quad (\text{B.4})$$

Algorithm 2: Dantzig Wolfe Decomposition Algorithm

Inputs : $\mathcal{G}$ Initial set of columns;

$\mathcal{J}$ Counter of dual variables;

$\mathcal{D}_i$ Set of optimal dual variables;

Outputs : D_i, F_1, F_2

1 Initialization

2 Solve Lp-Relaxation of *RMP* to obtain $\mathcal{D}_i$;

3 **if** $F_1 \leq F_2$ **then**

4 Add associated column F_2 ;

5 Go to Step 2 ;

6 **else**

7 Add associated column F_1 ;

8 Go to Step 2;

B.2. STAGE II: LP relaxation

In integer programming models, integer constraints may form a polytope (Raidl and Puchinger, 2008). Indeed, feasible space is not convex and the optimal solution is found at the extreme point of the convex hull of all feasible integer points (Dey et al., 2013). Therefore, the LP-Relaxation method (Desrosiers and Lübbecke, 2010) is applied to make feasible space convex to find an optimal solution. The LP-Relaxation (*LPR*) of the *RMP* can be given as:

$$\text{Min } LRRMP = \sum_{j_1=1}^{k_1} \eta_{j_1}^1 + \sum_{j_2=1}^{k_2} \eta_{j_2}^2 \quad (\text{B.5})$$

Subject to:

Constraints (B.2)–(B.3) &

$$\eta_{j_m}^m \in \mathbb{Z}^+ \quad \forall m \in \mathcal{M}; (\mathcal{M} = \{1, \dots, 8\}), j \in \{1, \dots, N\} \quad (\text{B.6})$$

B.2.1. Dual relaxed Restricted Master Problem

The dual formulation of *LRRMP* is given as follows:

$$\text{Min } DRRMP = \sum_{i \in \mathcal{J}} \mu_i \quad (\text{B.7})$$

Subject to:

$$\sum_{i=1,9} \mu_i \leq 1 \quad (\text{B.8})$$

$$\sum_{i=2,9} \mu_i \leq 1 \quad (\text{B.9})$$

$$\sum_{i=3,9} \mu_i = 0 \quad (\text{B.10})$$

$$\sum_{i=4,9} \mu_i = 0 \quad (\text{B.11})$$

$$\sum_{i=5,9} \mu_i = 0 \quad (\text{B.12})$$

$$\sum_{i=7,9} \mu_i = 0 \quad (\text{B.13})$$

$$\sum_{i \in \mathcal{I}} \mu_i \leq 1 \quad (\text{B.14})$$

$$\mu_i \in \mathbb{Z}^+ \quad \forall i = \{1, \dots, 5, 7, 8, 9\} \quad (\text{B.15})$$

Where μ_i represents dual variables.

B.3. STAGE III: Column generation (pricing problems)

The CG approach has been verified to be a desired and successful method in dealing with large-scale integer programs in recent decades (Wilhelm, 2001). The CG algorithm has been successfully implemented for various well-known problems, such as vehicle routing problems, crew scheduling, cutting stock problems, etc (Barnhart et al., 1998). One of the key advantages of the CG algorithm is that there is no need to enumerate possibilities (Ford and Fulkerson, 1958). In this method, n sub-problems should be solved in each iteration, and then variables associated with the most negative reduced cost (in maximisation, negative reduced cost and in minimisation, positive reduced cost) ($Z_j - C_j$) are selected to add to the initial columns (here since our problem is maximisation, so the most negative column should be selected). If there is no column to add to the initial columns, then the negative reduced cost matrix is empty, $(Z_j - C_j) = 0$ or $(Z_j - C_j) \geq 0$, and the optimal solution of the original problem is found.

B.3.1. First sub-problem

The first sub-problem (pricing) can be formulated as:

$$\begin{aligned} \text{Min } SP_1 = & [\gamma_9 \alpha_1 - \psi_1] \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} \sum_{t \in \mathcal{T}} X_{po'd't} \\ & + [\gamma_9 \alpha_2 - \psi_2] \sum_{p \in \mathcal{P}} \sum_{o' \in \mathcal{O}'} \sum_{d' \in \mathcal{D}'} \sum_{t \in \mathcal{T}} X'_{po'd't} \\ & + [\gamma_9 \alpha_3 - \psi_3] \sum_{i \in \mathcal{I}} \bar{X}_i \\ & + [\gamma_9 \alpha_4 - \psi_4] \sum_{p \in \mathcal{P}} Z_p + [\gamma_9 \alpha_5 - \psi_5] \sum_{g \in \mathcal{G}} Y_g \\ & + [\gamma_9 \alpha_6 - \psi_6] \sum_{g' \in \mathcal{G}'} Y'_{g'} + \sum_{i=\{1, \dots, 6\}} \mu_i \end{aligned} \quad (\text{B.16})$$

Subject to:

Constraints (2)–(9), (15), (17)–(20) &

$$\mu_i \in \mathbb{Z}^+ \quad \forall i = \{1, \dots, 6\} \quad (\text{B.17})$$

Where μ_i are dual variables, $[\gamma_9 \alpha_1 - \psi_1]$ calculates the coefficient of variables $(Z_j - C_j)$ where C_j represents the negative reduced cost.

B.3.2. Second sub-problem

The second sub-problem (pricing) can be written as:

$$\text{Min } SP_2 = (\gamma_9 \alpha_7 - \psi_7) \sum_{v \in \mathcal{V}'} \sum_{o' \in \mathcal{O}'} \sum_{d \in \mathcal{N}} \bar{Y}_{vo'd} + \sum_{i=\{7,8\}} \mu_i \quad (\text{B.18})$$

Subject to:

Constraints (10)–(14) &

$$\mu_i \in \mathbb{Z}^+ \quad \forall i = 7, 8 \quad (\text{B.19})$$

Where μ_i are dual variables, $(\gamma_9\alpha_7 - \psi_7)$ calculates the coefficient of the variable as objective function ($Z_j - C_j$ (Negative reduced cost)). The pseudo-code of CG algorithm is briefed as follows:

Algorithm 3: Column Generation Algorithm

Inputs : $\mathcal{A}$ Set of feasible columns;
 $\mathcal{D}_i$ Set of optimal dual variables;
 $\mathcal{J}$ Set of initial columns;
Parameters: N_c Negative reduced costs matrix;
 cc Column with the most negative reduced cost;
Outputs : D_i, obj_1, obj_2

```

1 while  $N_c \neq \{\emptyset\}$  do
2   Solve  $RMP$ ;
3   Define  $obj_1$  and  $obj_2$  using dual values of  $LRRMP$ ;
4   Solve  $SP_1, SP_2$  to find  $N_c$ ;
5   if  $N_c \neq \{\emptyset\}$  then
6      $\mathcal{J} \leftarrow cc$ ;
7   else
8     Go to Step 2;
```

B.3.3. STAGE IV: Dominance based neighborhood search algorithm (DBNSA)

In this section, an algorithm named DBNSA is designed to add columns with negative reduced costs to the $LRRMP$. In other words, this algorithm is designed to integrate into the CG algorithm (Developed CG) in order to find columns with negatively reduced cost by solving pricing problems and adding those columns to the Restricted Master Problem. First, the LP-Relaxation of the Restricted Master Problem should be solved to find a good bound. Then, the pricing problems should be solved. After that, the swap fabrication operator produces its neighbours, if the objective function value is worse than obtained bound, this solution will be discarded, otherwise, the algorithm will continue with the best-found solution in the neighbours, and will continue until there is no column to add to the Restricted Master Problem. In order to prevent trapping in the local optimum, the algorithm finds the feasible solution's neighbours, which leads to diversification in the algorithm to find the best optimal solution. Finally, the column associated with the best-found solution is given to the master problem.

Algorithm 4: Dominance Based Neighborhood Search Algorithm

Inputs : $\mathcal{A}$ Set of feasible columns
 $\mathcal{D}_i$ Set of optimal dual variables
Parameters: S_a Set of all neighbors of feasible solution a ;
 FO_a Objective value of each feasible solution;
 LB Lower bound;
 B Best solution;
 N_c Negative reduced costs matrix;
Outputs : N_c, B

```

1 Solve  $LRRMP$  (LP Relaxed Master Problem)
2 Find  $S_a$  by swap operator;
3 if  $FO_a \leq LB$  then
4   #Abort;
5 else if  $FO_a \geq FO_{a+1}$  then
6    $B = FO_{a+1}$ ;
7 else if  $FO_a \leq FO_{a+1}$  then
8    $B = FO_a$ ;
```

B.3.4. Branching strategy

In this section, the branching strategy is utilised to extend the $B\&P$ tree. If the obtained solutions (variables $X_{po'd't}$, $X'_{po'd't}$ and $\bar{x}_i$) are not integers, two branches will be expanded and the problem should be solved with new branches individually. The algorithm does not stop as long as finds integer values for these variables. The branches will be generated as follows:

$$X_{po'd't} \geq \lfloor X_{po'd't} \rfloor \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T} \quad (B.20)$$

$$X_{po'd't} \leq \lceil X_{po'd't} \rceil \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T} \quad (B.21)$$

$$X'_{po'd't} \geq \lfloor X'_{po'd't} \rfloor \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T} \quad (B.22)$$

$$X'_{po'd't} \leq \lceil X'_{po'd't} \rceil \quad \forall p \in \mathcal{P}, o' \in \mathcal{O}', d' \in \mathcal{D}', t \in \mathcal{T} \quad (B.23)$$

$$\bar{X}_i \geq \lfloor \bar{X}_i \rfloor \quad \forall i \in \mathcal{I} \quad (B.24)$$

$$\bar{X}_i \leq \lceil \bar{X}_i \rceil \quad \forall i \in \mathcal{I} \quad (B.25)$$

As mentioned above, the $LRRMP$ provides a proper bound, therefore, this can be used to construct a branching rule. If the branches' value exceeds the obtained bound from the $LRRMP$, the branches that exceed the bound can be cut. Consequently, the convergence speed will be improved by using the above-mentioned branching rules.

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